

DATE : 24.07.2025

DT/NT : DT

LESSON : DEEP LEARNING

**SUBJECT: INTRODUCTION TO CNN &
COMPUTER VISION**

BATCH : B287

**DATA
SCIENCE**



TECHPRO
EDUCATION



techproeducation.com



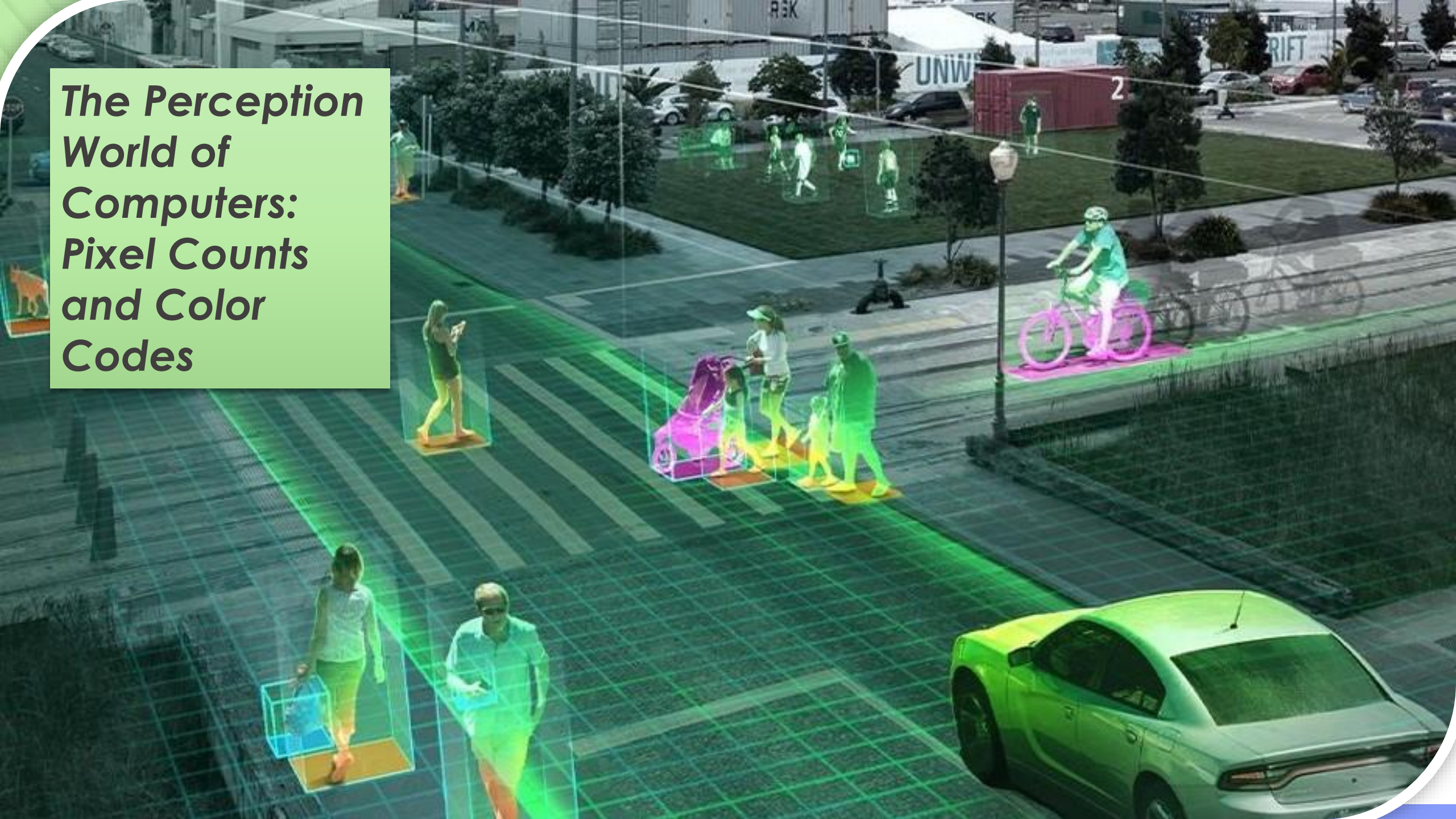
+1 (585) 304 29 59





01
01

The Perception World of Computers: Pixel Counts and Color Codes



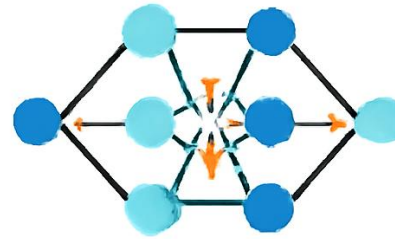


General Overview

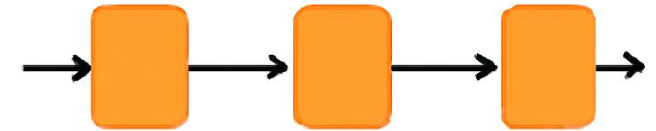
Contents in Deep Learning

- ✓ ANN
- ✓ RNN-LSTM-GRU
- ✓ NLP-LLM
- ✓ CNN-CV

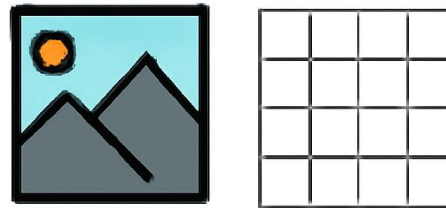
ANN



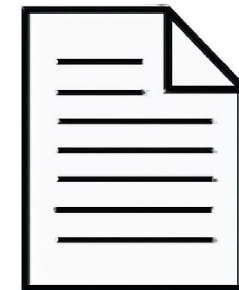
RNN - LSTM - GRU



CNN - CV



NLP

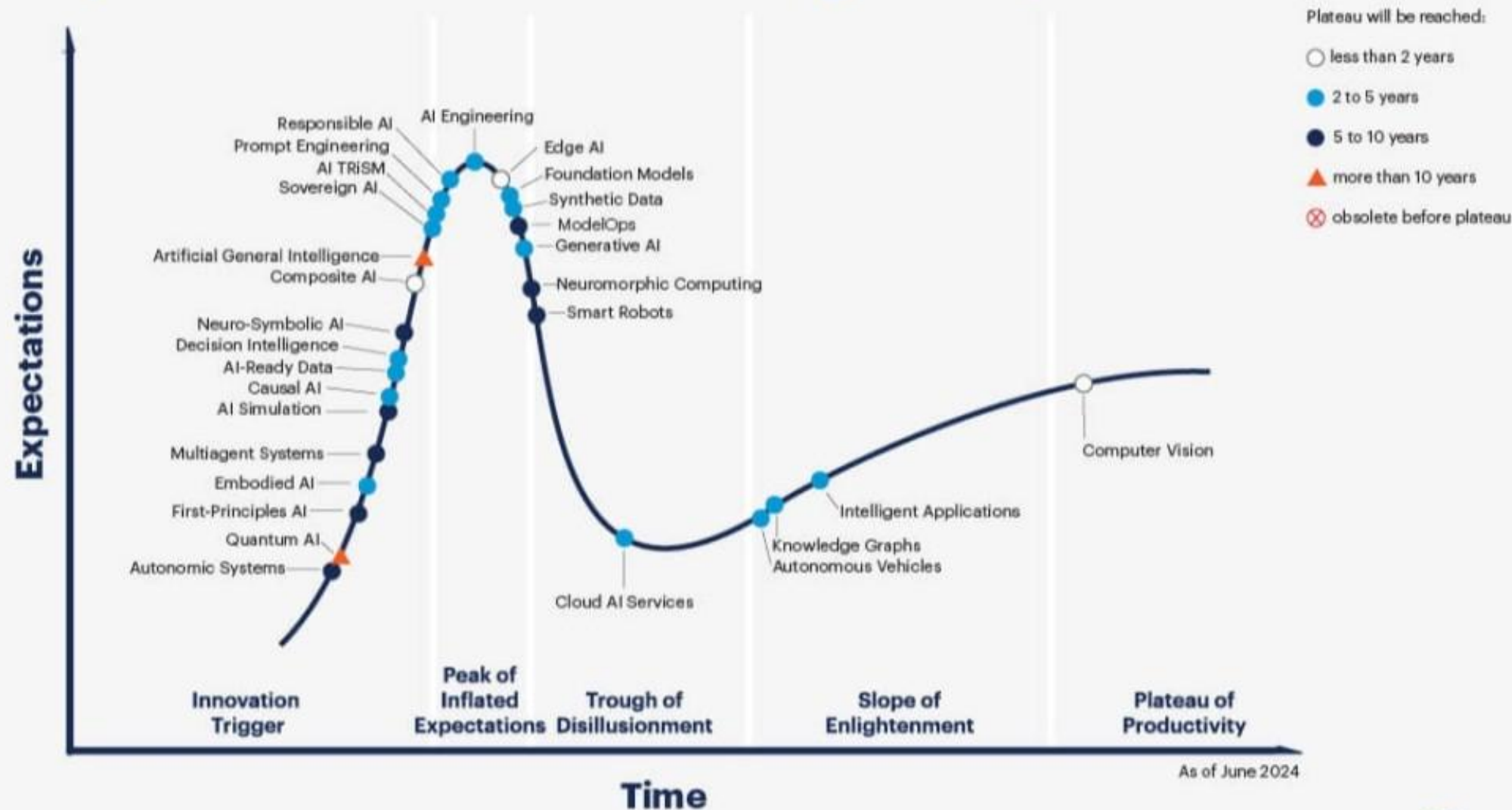




Contents

- ✓ **The Perception World of Computers: Pixel Counts and Color Codes**
- ✓ **CNN (Convolutional Neural Network) and Computer Vision**
- ✓ **Transfer Learning**
- ✓ **Most Popular Python Libraries for Computer Vision/Image Processing**
 - ☐ **Pillow**
 - ☐ **Scikit-image**
 - ☐ **OpenCV**
- ✓ **Labeling (Makesense/Roboflow)**
- ✓ **Yolo**

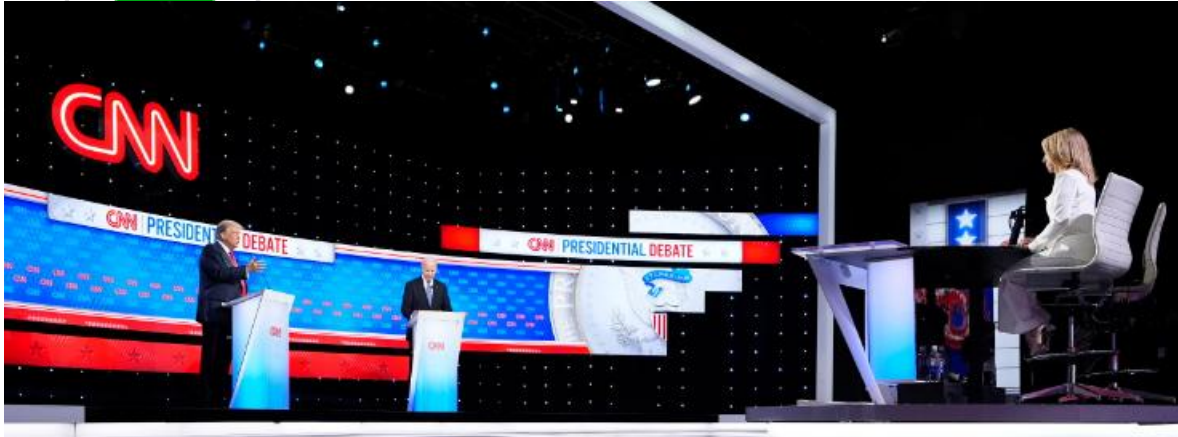
Hype Cycle for Artificial Intelligence, 2024



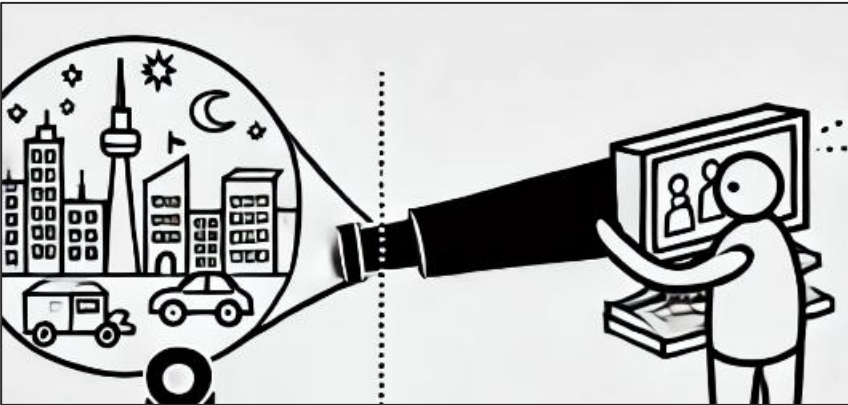
Source: Gartner
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© 2024 Gartner, Inc. and/or its affiliates. All rights reserved. GTS_3282450



Convolutional Neural Network (CNN) and Computer Vision (CV)



When you see CNN, remember it is a TV channel, and it relates to images.



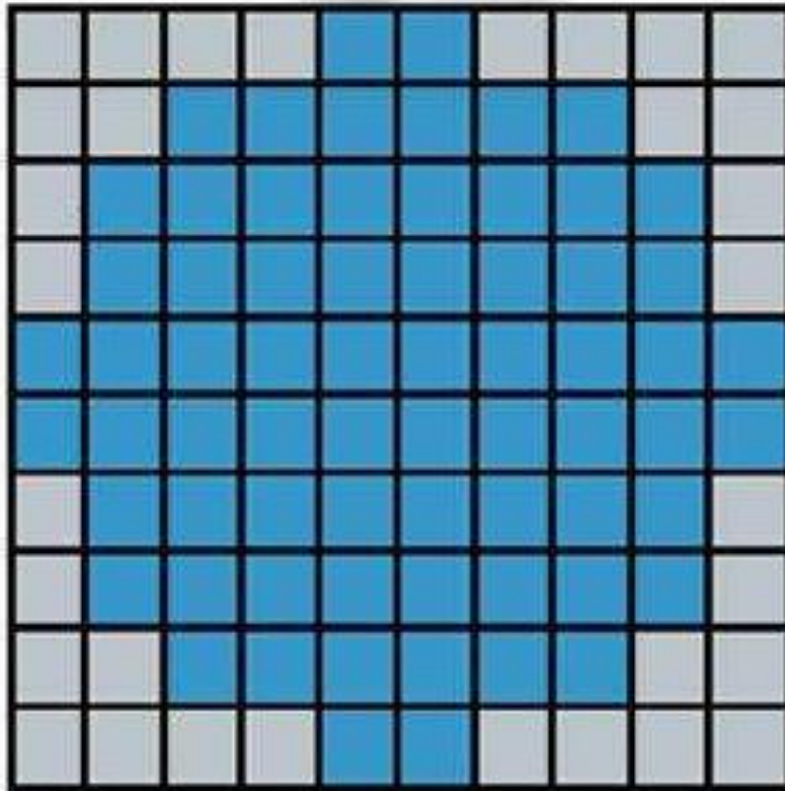
We are trying to teach computers how they can see and understand images.





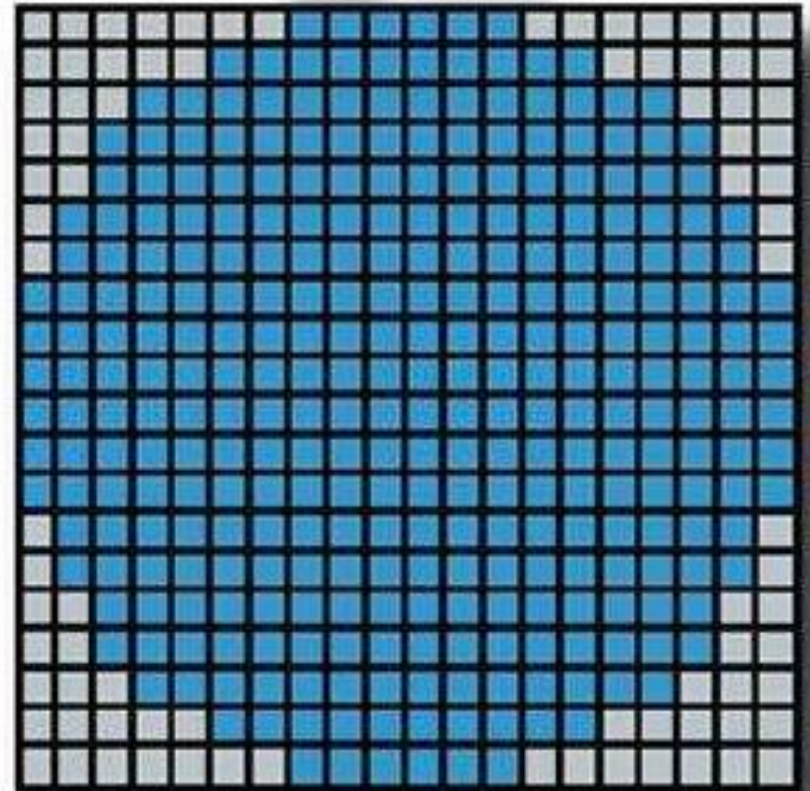
CNN and Computer Vision

10 PPI



2,54 cm

20 PPI

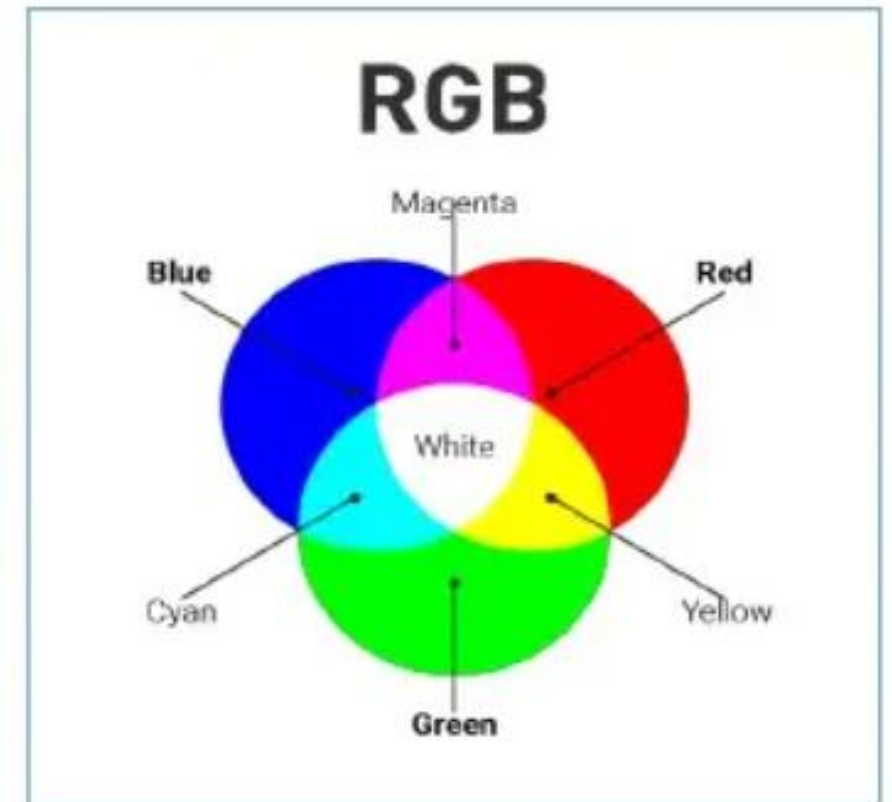


2,54 cm



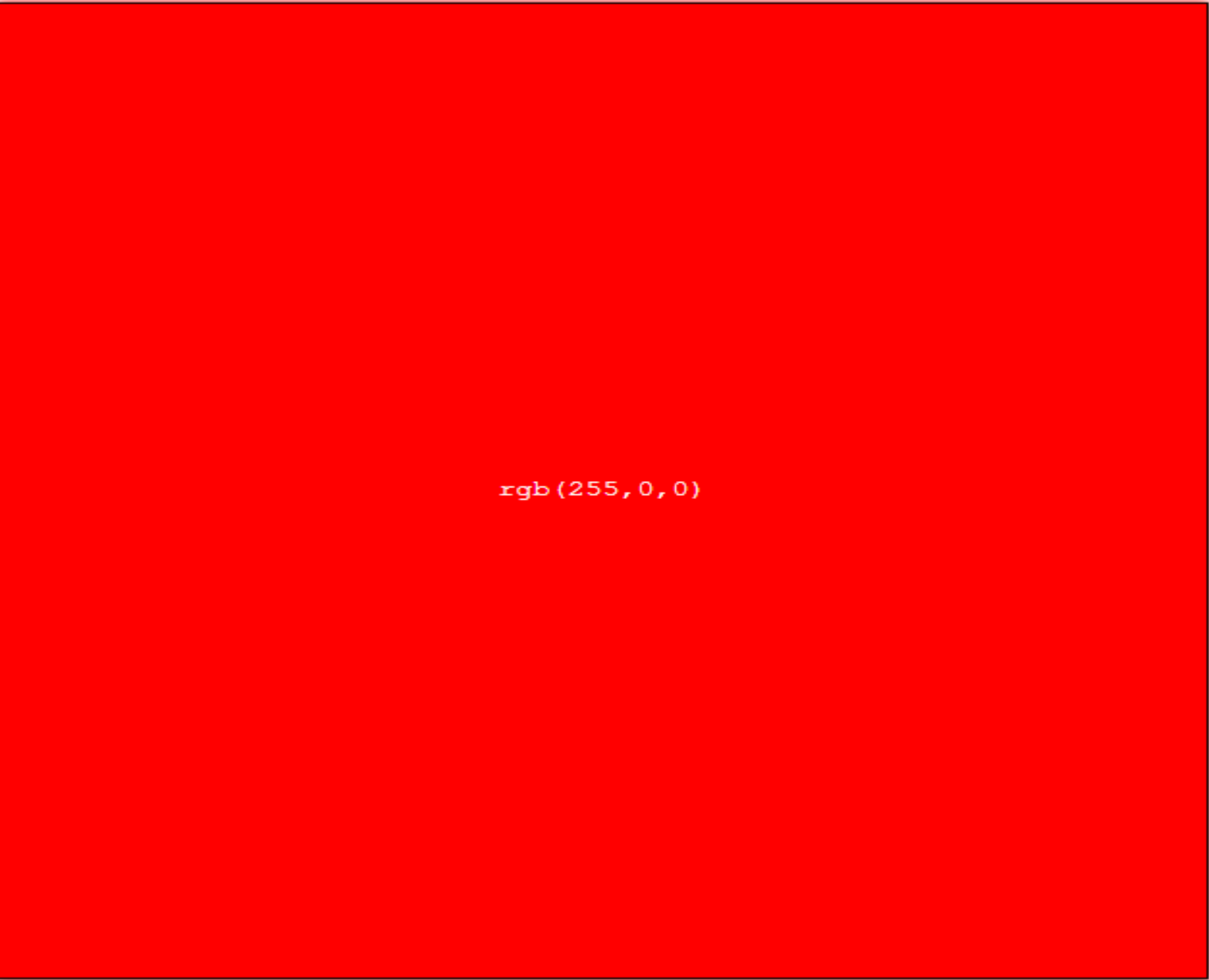
CNN and Computer Vision

Color Name	RGB Value (0 to 255)			Color
	B (Blue)	G (Green)	R (Red)	
Blue	255	0	0	
Green	0	255	0	
Red	0	0	255	
White	255	255	255	
Black	0	0	0	



https://www.youtube.com/shorts/Z6SA_I8KXNU

RGB Colour Mixer - Used by Screens



Red - Currently set to 255

- 0
- 32
- 64
- 96
- 128
- 159
- 191
- 223
- 255

Green - Currently set to 0

- 0
- 32
- 64
- 96
- 128
- 159
- 191
- 223
- 255

Blue - Currently set to 0

- 0
- 32
- 64
- 96
- 128
- 159
- 191
- 223
- 255



How to See??

Camera



Lighting



**Vision
Software**

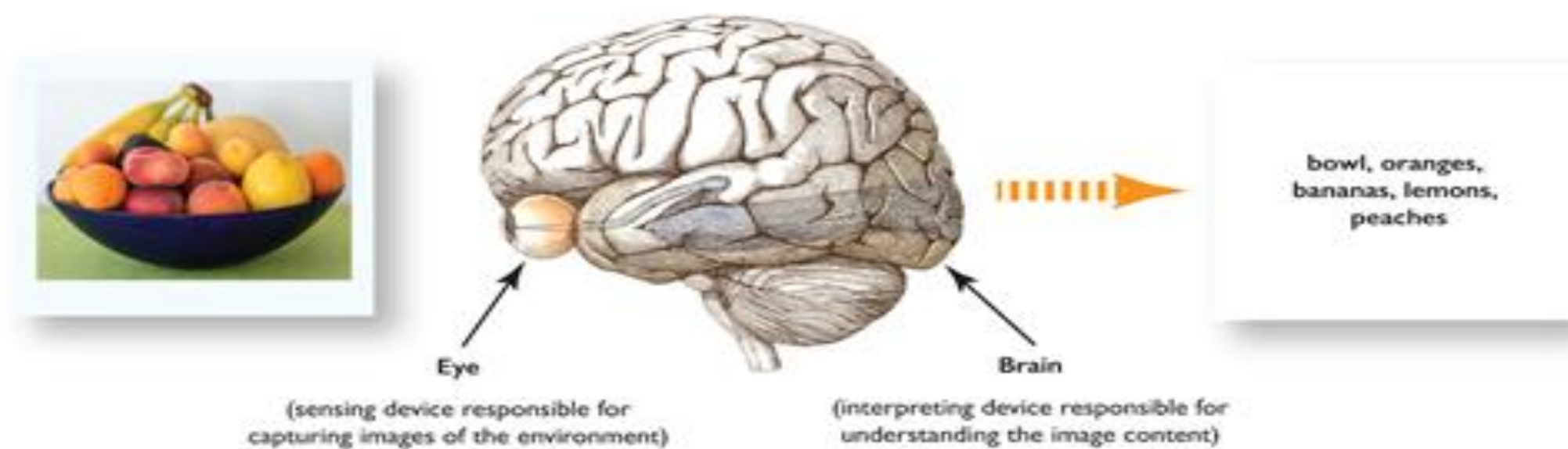


Scene Description



Scene

Human Vision System



Computer Vision System





CNN and Computer Vision



01
01



CNN and Computer Vision





CNN and Computer Vision





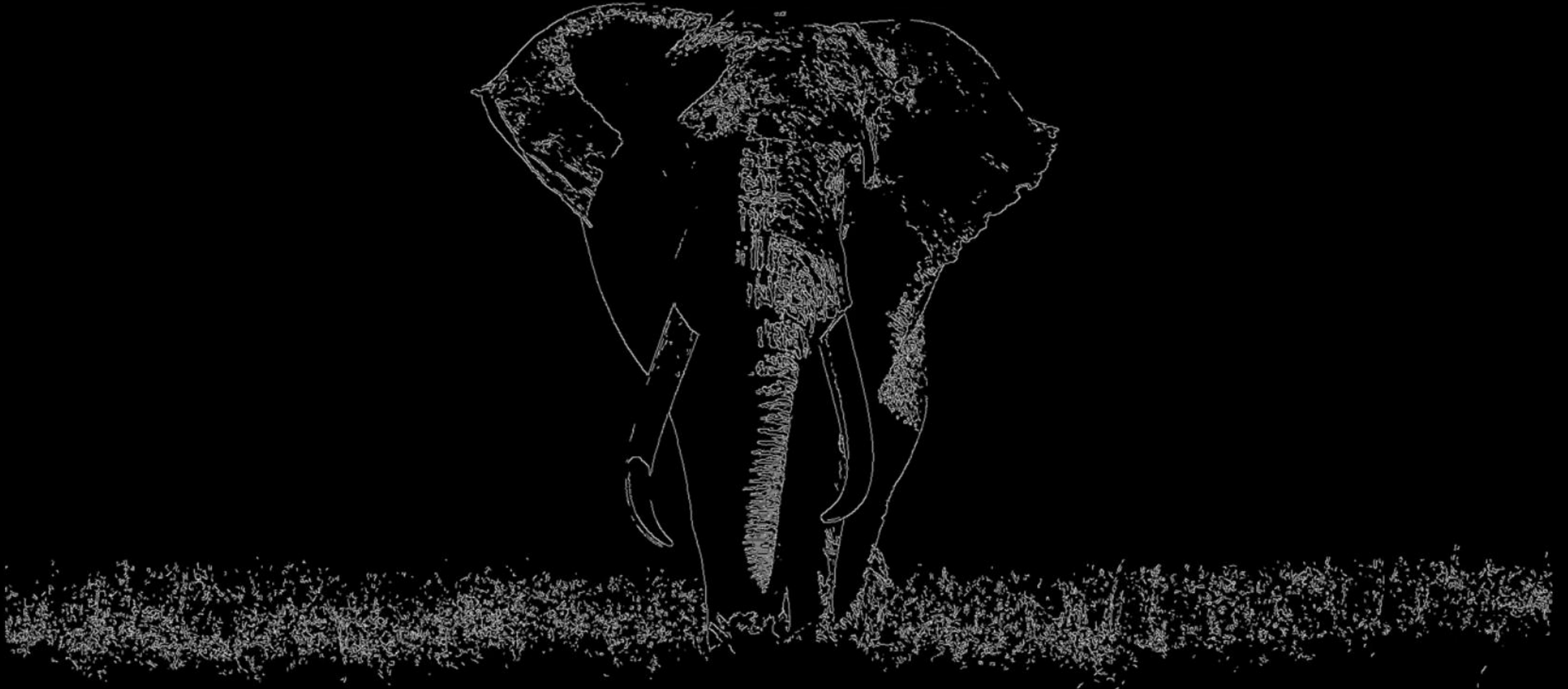
CNN and Computer Vision



01
01



CNN and Computer Vision





Handwritten Digit Recognition

- Yann LeCun, Corinna Cortes and Christopher J.C. Burges (1989) developed first Convolutional Network for Handwritten Digit Recognition.

VIDEO:

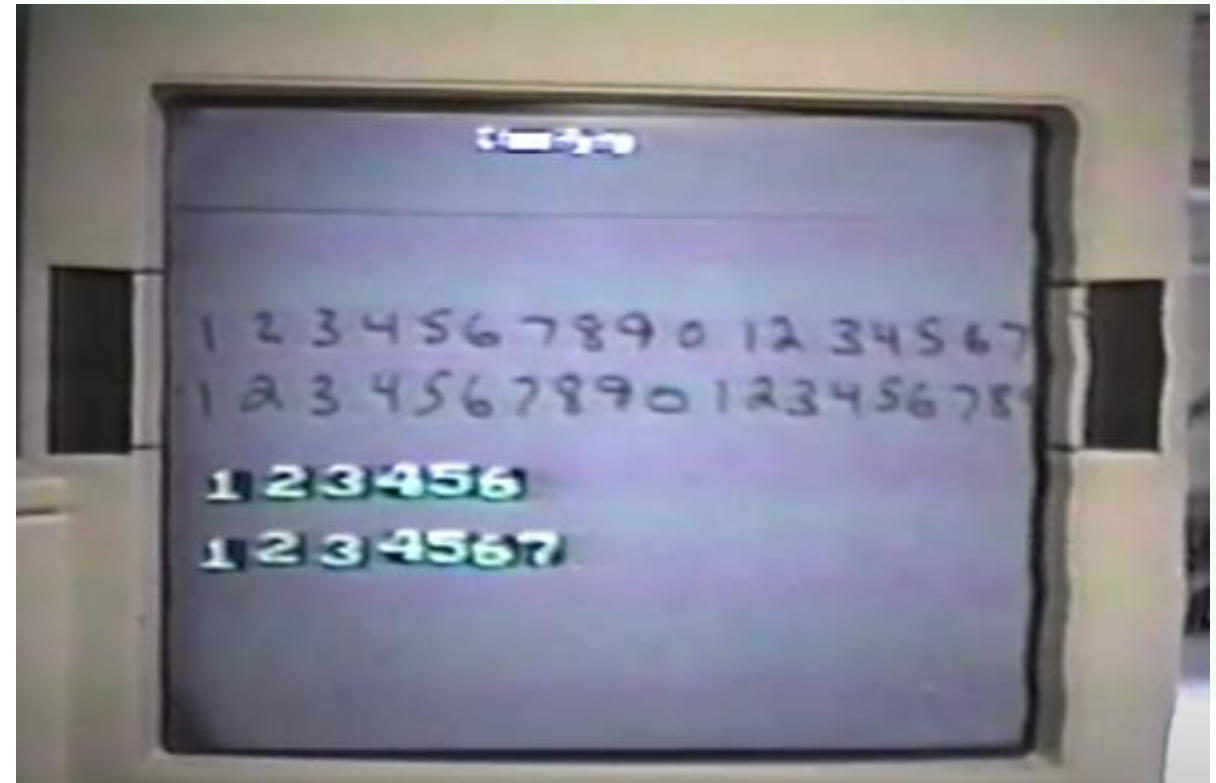
https://www.youtube.com/watch?v=FwFduRA_L6Q

ARTICLE: chrome-

[extension://efaidnbmnnnibpcajpcglclefindmkaj/https://proceedings.neurips.cc/paper/1989/file/53c3bce66e43be4f209556518c2fcb54-Paper.pdf](https://proceedings.neurips.cc/paper/1989/file/53c3bce66e43be4f209556518c2fcb54-Paper.pdf)

DL: <https://www.nature.com/articles/nature14539>

- 2015



Fei-Fei Li:

How we're teaching computers to understand pictures

TED2015 · 17:58 · **Filmed** Mar 2015

Subtitles available in 15 languages

 [View interactive transcript](#)

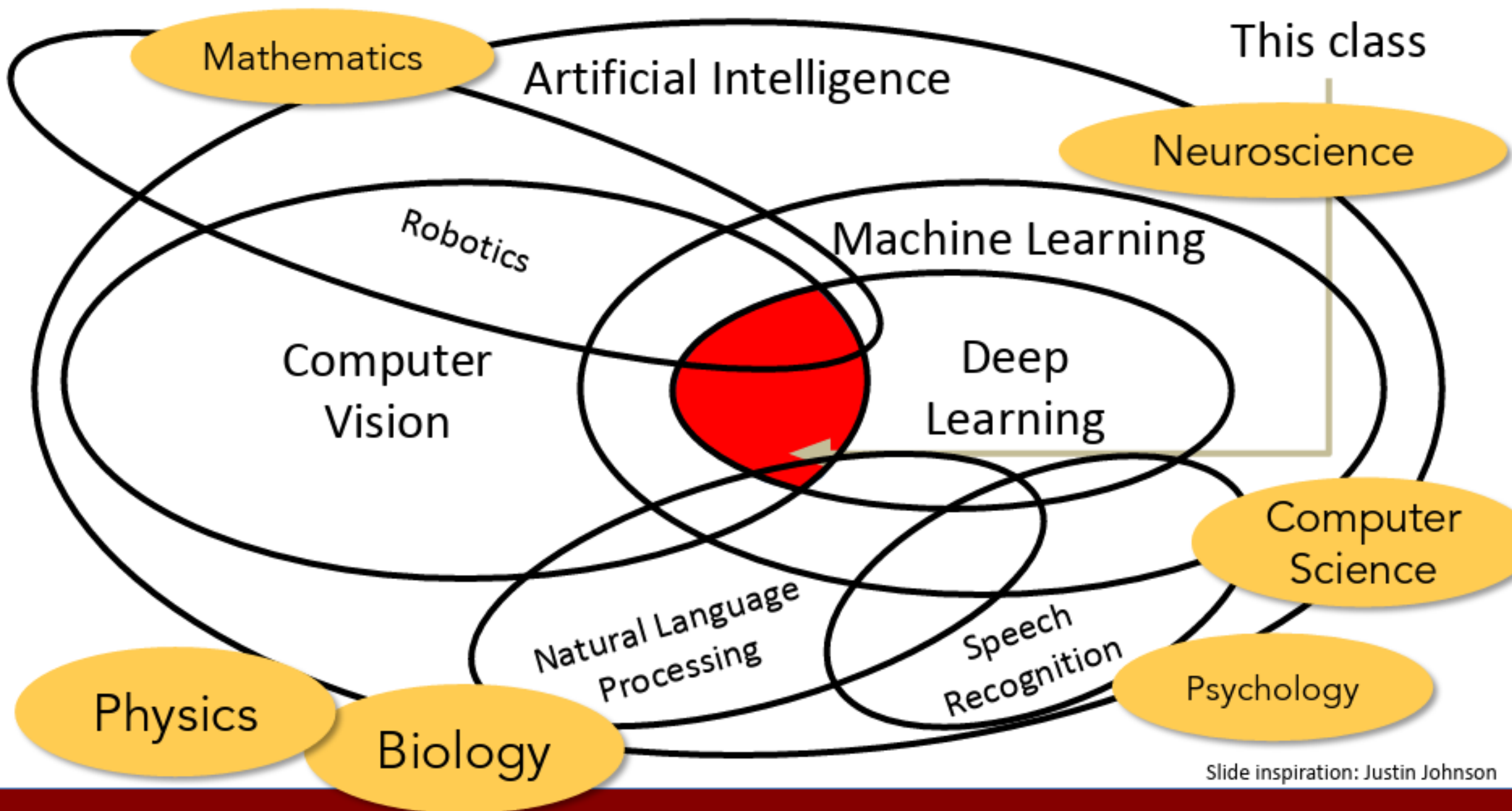




Two Big AI Gru in a Short Video



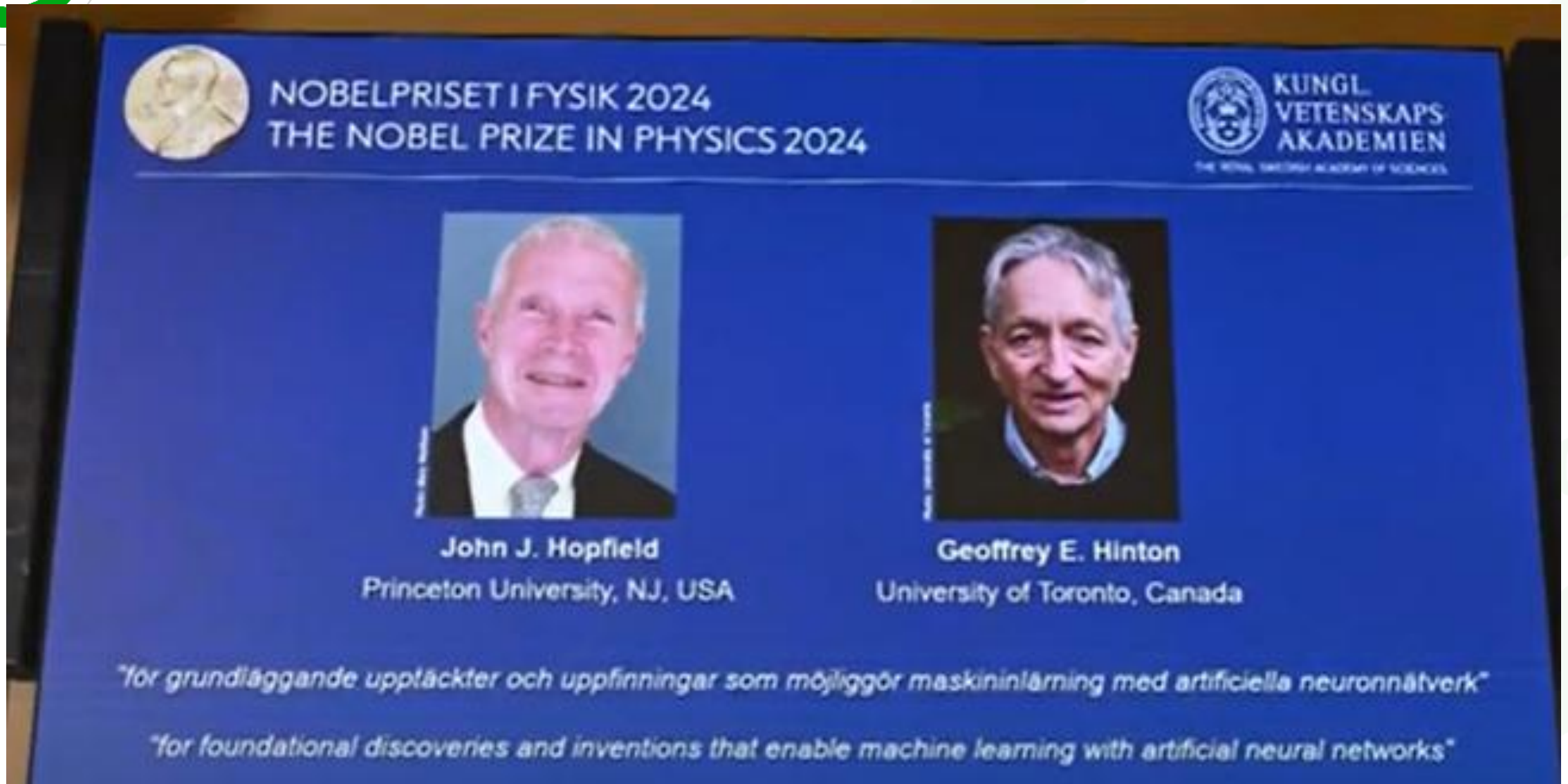
<https://www.youtube.com/shorts/cxJi15eXWJU>



Slide inspiration: Justin Johnson



Nobel Prize Come to AI Technologies





Nobel Prize Come to AI Technologies

University of Glasgow

→ STUDY → RESEARCH → EXPLORE → CO

UNIVERSITY NEWS

PROFESSOR DARON ACEMOGLU LAUNCHES ADAM SMITH DISTINGUISHED SPEAKER SERIES

Published: 16 May 2024

Globally renowned political economist Professor Daron Acemoglu delivered an inaugural lecture on Artificial Intelligence (AI) and its role in the economy and society at the University of Glasgow.



EKONOMIPRISET 2024
THE PRIZE IN ECONOMIC SCIENCES 2024

KUNGL. VETENSKAPS AKADEMIEN
THE ROYAL SWEDISH ACADEMY OF SCIENCES

 Daron Acemoglu Massachusetts Institute of Technology, USA	 Simon Johnson Massachusetts Institute of Technology, USA	 James A. Robinson University of Chicago, IL, USA
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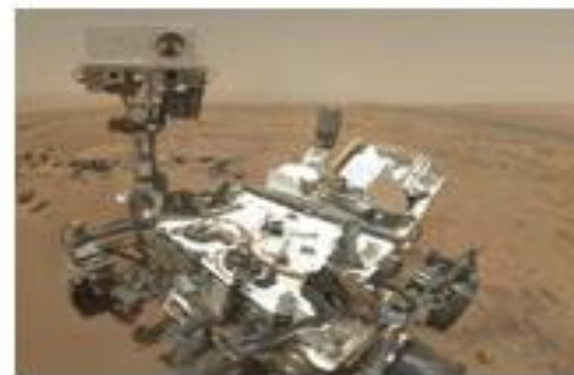
"för studier av hur institutioner formas och påverkar välbefinnande"
"for studies of how institutions are formed and affect prosperity"





What is the role of computer vision in our daily lives?

Computer Vision is everywhere!



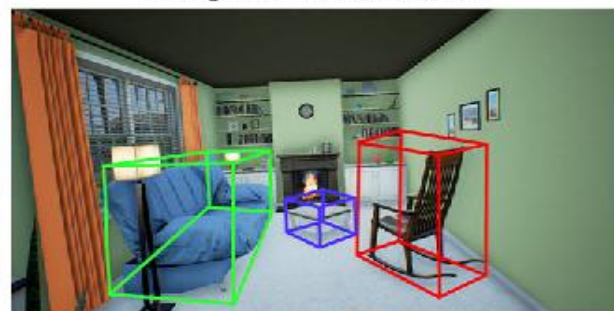
Object Tracking



Pose Estimation



Object Detection



Action Recognition



Autonomous Navigation



3D Reconstruction



Crowd Understanding



Urban Scene Understanding



Indoor Scene Understanding



Multi-agent Collaboration



Human Training



Aerial Surveying



Red Image
Green Image Label

Orange Depth/Multi-View
Yellow User Input

Purple Video
Brown Physics

Pink Segmentation/Bounding Box
Gray Camera Localization



Real-Time Sign Language Detection for Numbers (0-9)



https://www.linkedin.com/feed/update/urn:li:activity:7255810674545635328?utm_source=share&utm_medium=member_desktop



https://www.linkedin.com/feed/update/urn:li:activity:7256186549426429954?utm_source=share&utm_medium=member_desktop

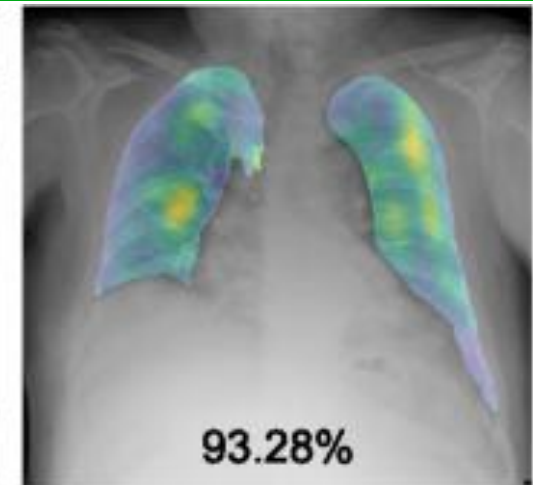
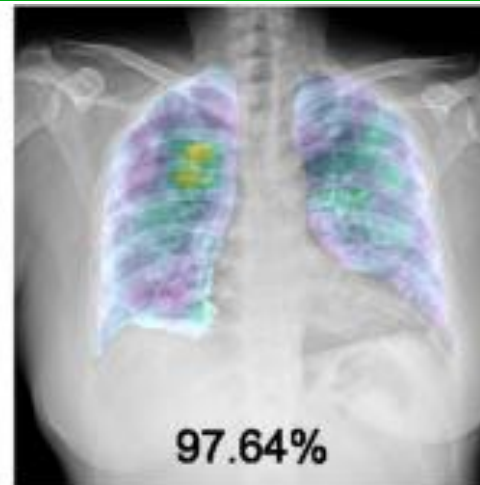
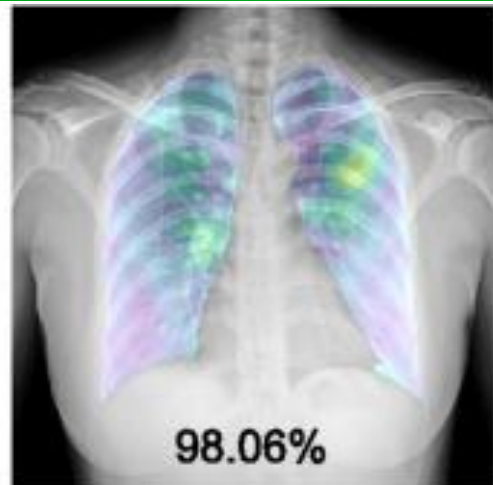
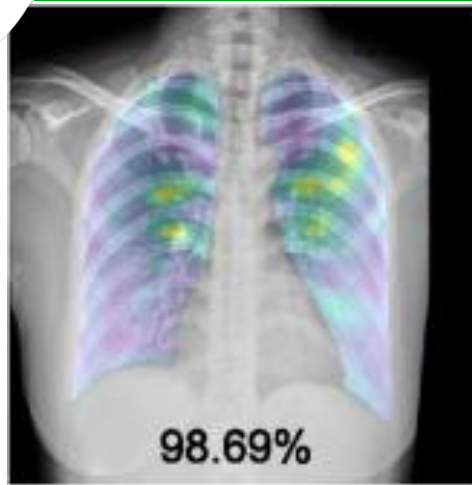
An idea for you



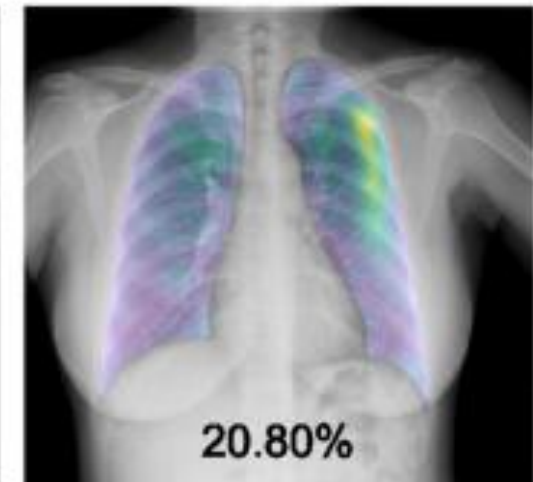
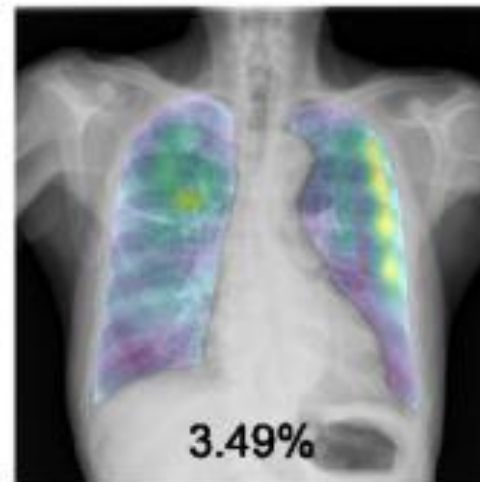
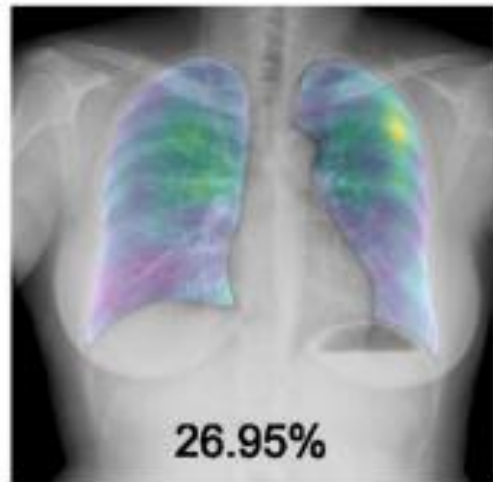
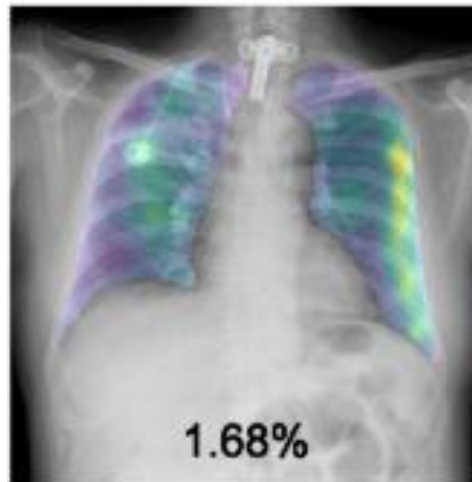
https://www.tiktok.com/@twin_kl_resources/video/7192623988244253957?lang=en



CNN and Computer Vision



(a)



(b)



Breast Cancers Detection in the UK by Using CV

More breast cancers detected in first evaluation of breast screening AI

21 March 2024

"It's a lifesaver, it's a life changer," says one of the first women in the UK to have her breast cancer picked up by AI software, Mia

An artificial intelligence (AI) breast screening solution called Mia helped doctors find an additional 12 percent more cancers than in routine practice, as announced today in collaboration with Kheiron Medical Technologies, NHS Grampian, the University



Funding award for new database to aid early diagnosis in breast cancer

26 March 2024

Local charity, Friends of ANCHOR is the key funder for a new digital dataset that will continue the drive to optimise artificial intelligence (AI) in diagnosing breast cancer in north-east Scotland.

Early diagnosis of breast cancer in north-east Scotland is about to take a significant step forward thanks to local charity funding.



Aberdeen AI trial helps doctors spot breast cancers

9 June 2023 · 36 Comments



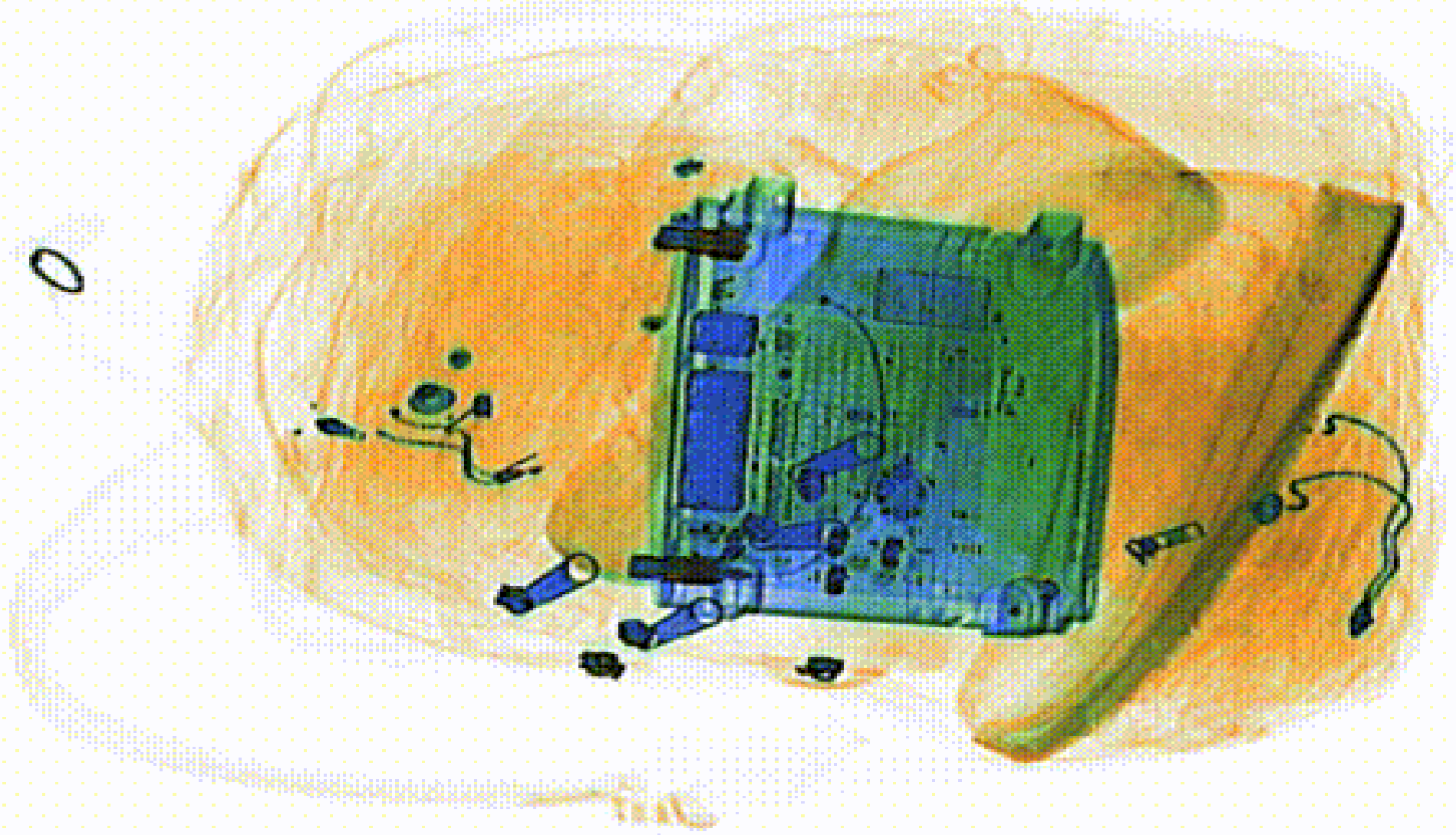
Mia in comparison to the first reader

• 1st reader (human)

- Recall rate : 5.7%
- Cancer detection rate: 5.4/1000
- Sensitivity: 86.1% (261 cancers)
- Specificity: 95.2%

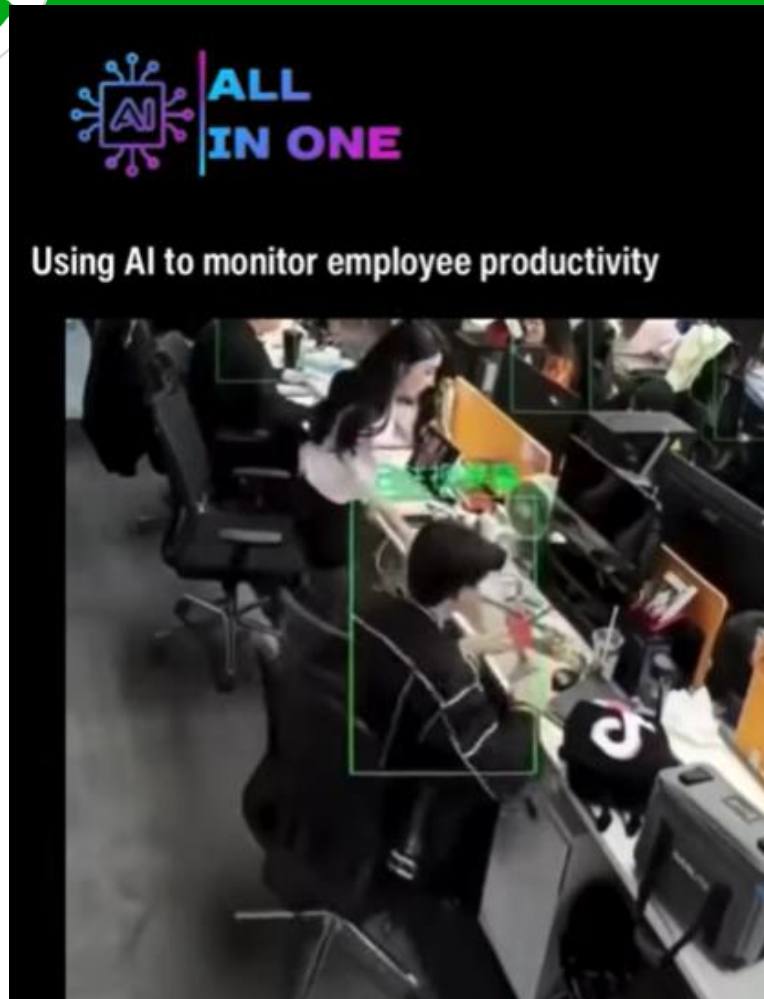
• Mia

- Recall rate : 13.0%
- Cancer detection rate: 6.1/1000
- Sensitivity: 91.4% (277 cancers)
- Specificity: 87.6%



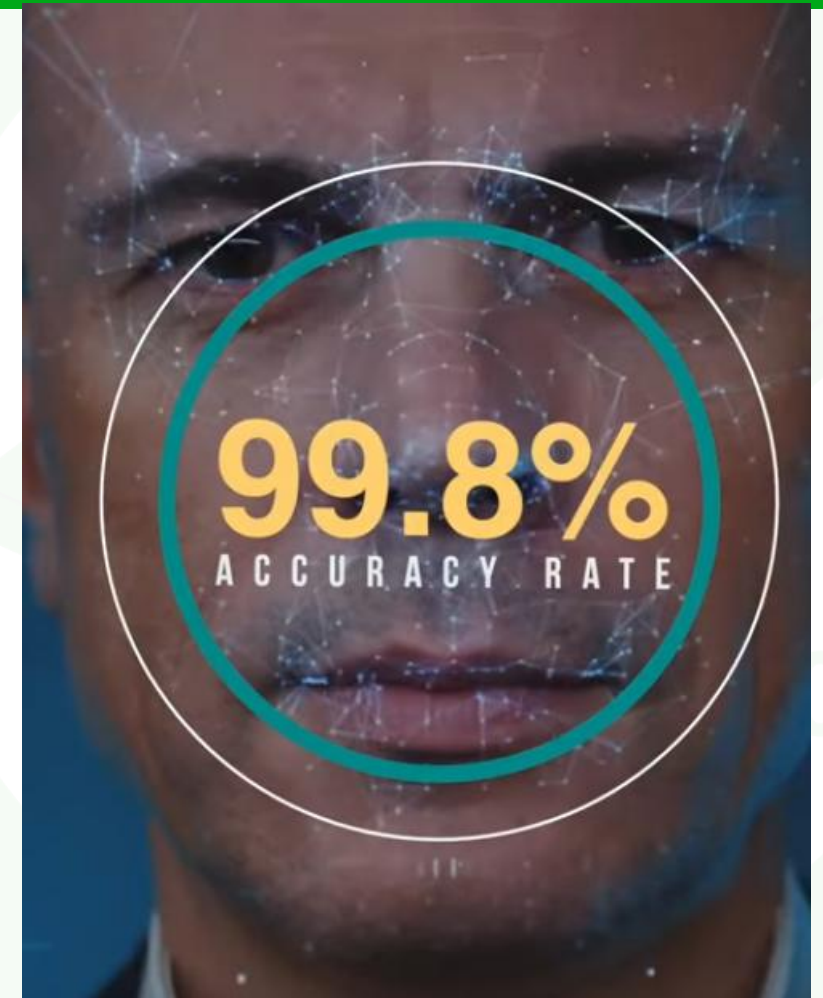


Employee Tracking & Human Recognition Systems



<https://www.youtube.com/shorts/QNIGILQ-syw>

https://www.linkedin.com/feed/update/urn:li:activity:7254055469671952385?utm_source=share&utm_medium=member_desktop



<https://www.youtube.com/shorts/36wo9CL7YJQ>



SELF-DRIVING CARS





In China

SELF-DRIVING CARS



https://www.youtube.com/watch?v=FQRe08_U60c

<https://www.youtube.com/watch?v=55jYthD1Wo4>

<https://www.youtube.com/shorts/mxxzscEDH1w> Horsecar



Greengrocer / Cofee Delivery Robots at Cambridge

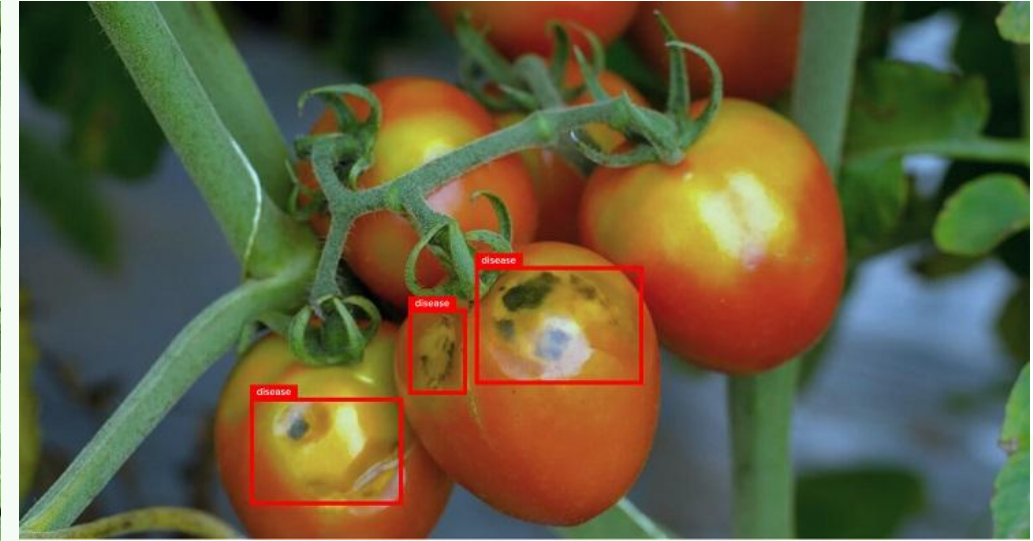
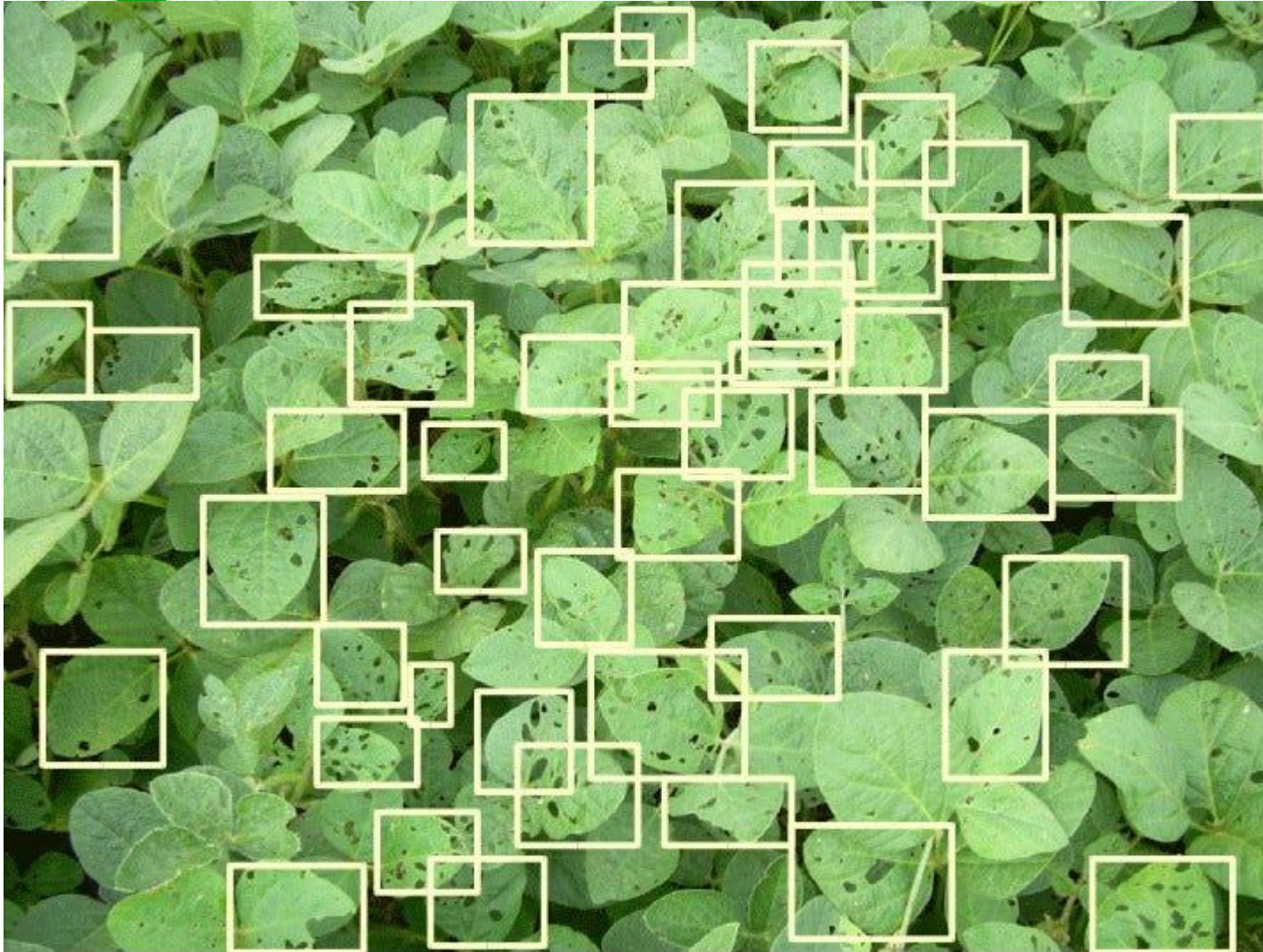


Robots serving your Starbucks cofee

<https://www.youtube.com/watch?v=WBHhjRSZPtU>

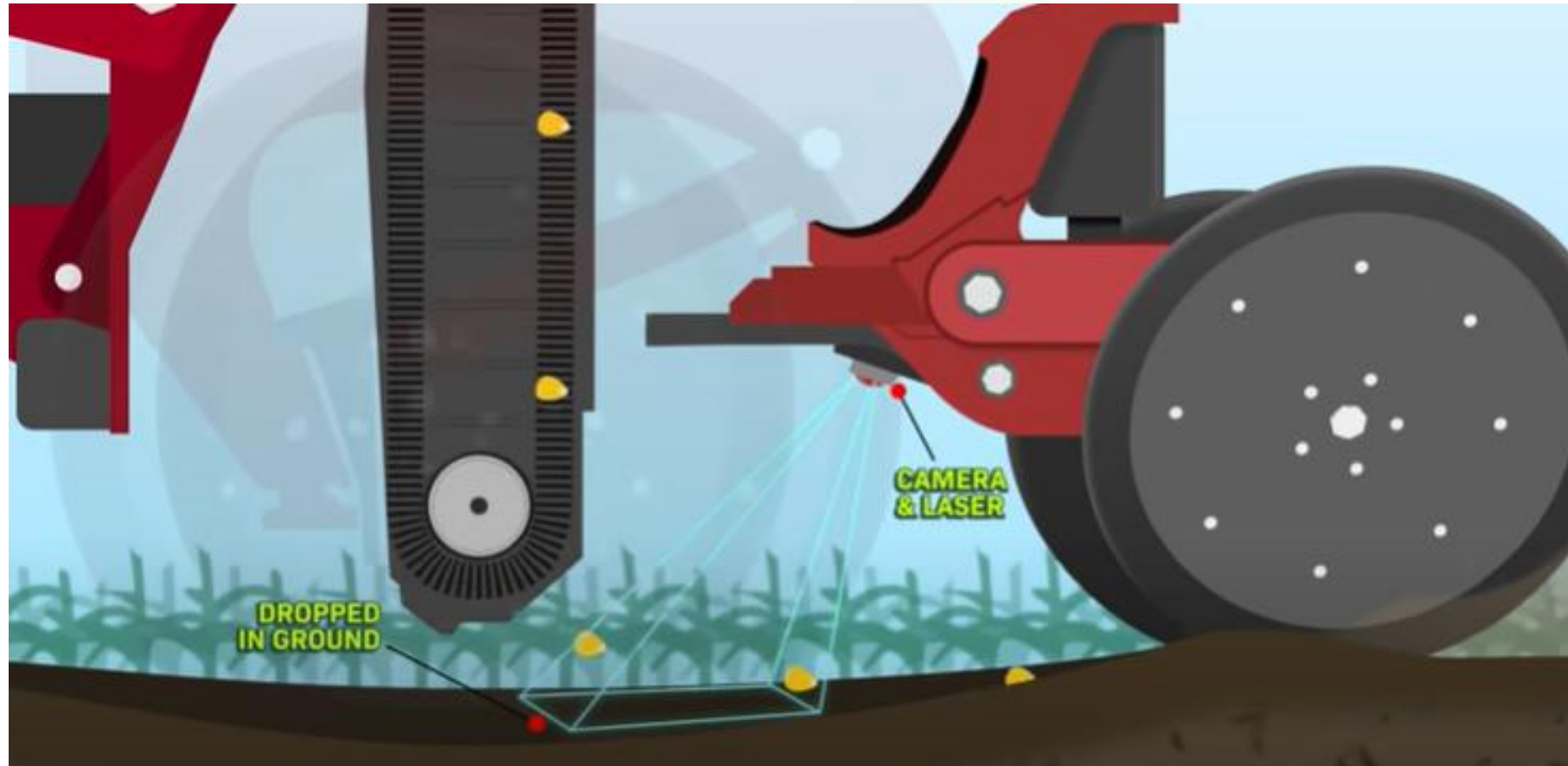


CNN and Computer Vision





Computer Vision in Farming Sector





WHAT COMPUTERS SEE



CNN and Computer Vision

What computers 'see': Images as Numbers

What you see



Input Image

What you both see

157	153	174	168	160	152	129	153	172	163	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

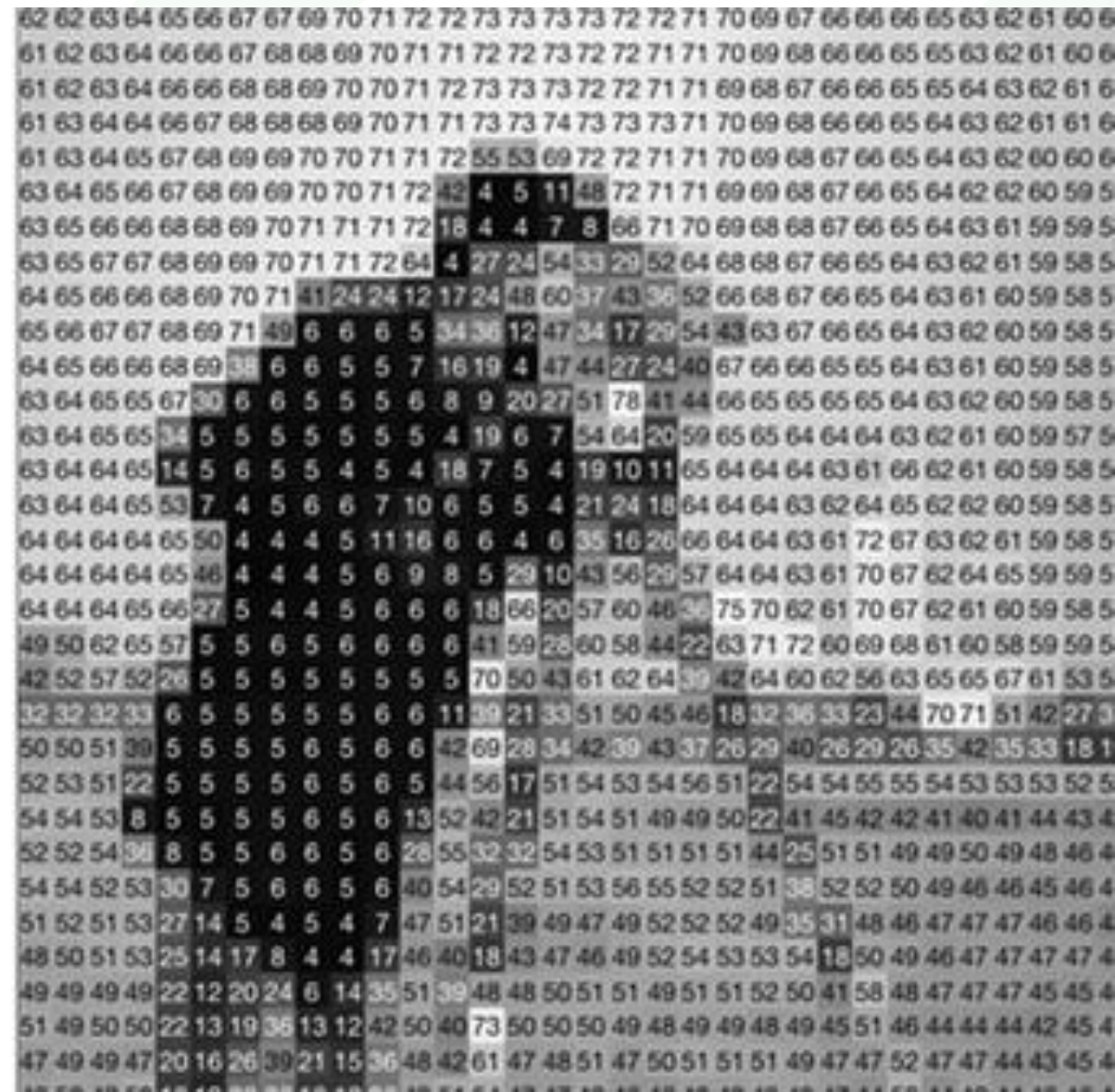
Input Image + values

What the computer "sees"

157	153	174	168	160	152	129	153	172	163	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	106	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

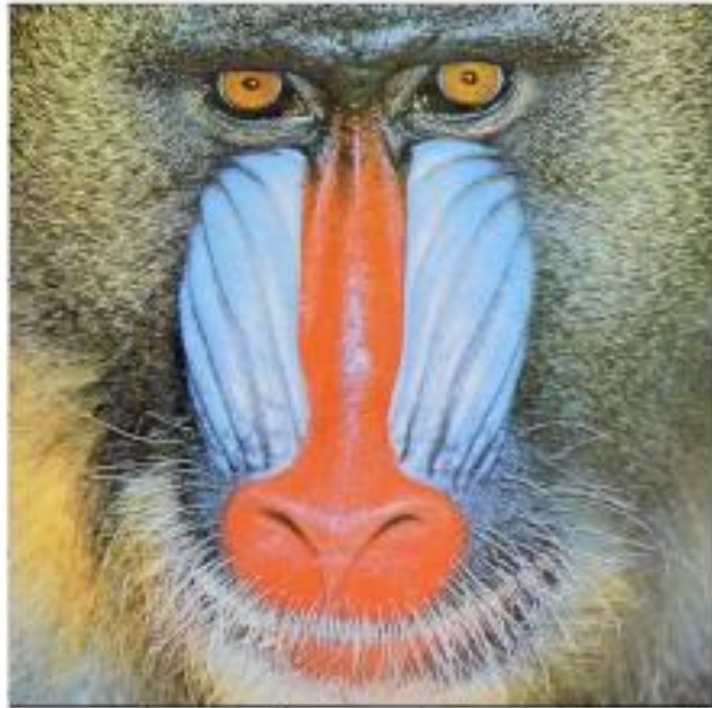
Pixel intensity values
("pix-el"=picture-element)

An image is just a matrix of numbers $[0,255]$. i.e., $1080 \times 1080 \times 3$ for an RGB image.



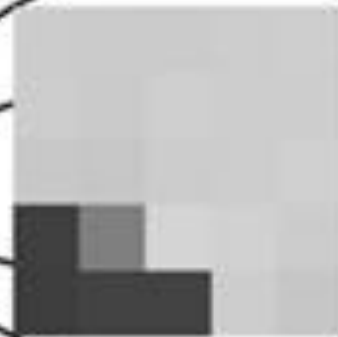


CNN and Computer Vision


$$\begin{bmatrix} 9 & 3 & \dots & 40 \\ 2 & 5 & 3 & \dots & 40 \\ 5 & 3 & 3 & \dots & 40 \\ 4 & 5 & 3 & 3 & \dots & 40 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 5 & 3 & 4 & \dots & 24 \end{bmatrix}$$



CNN and Computer Vision



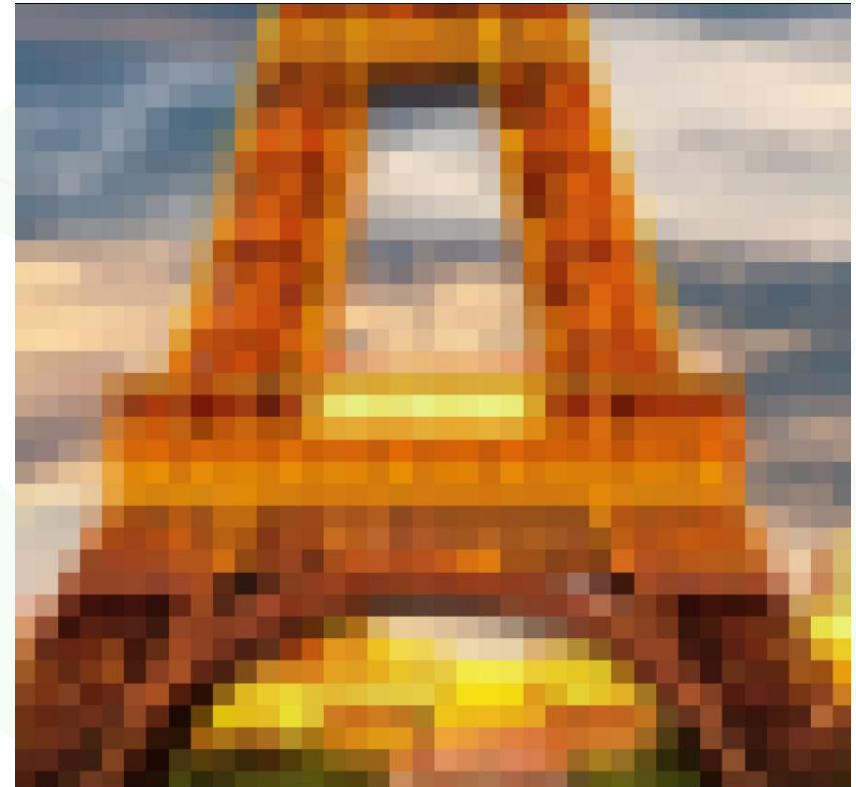
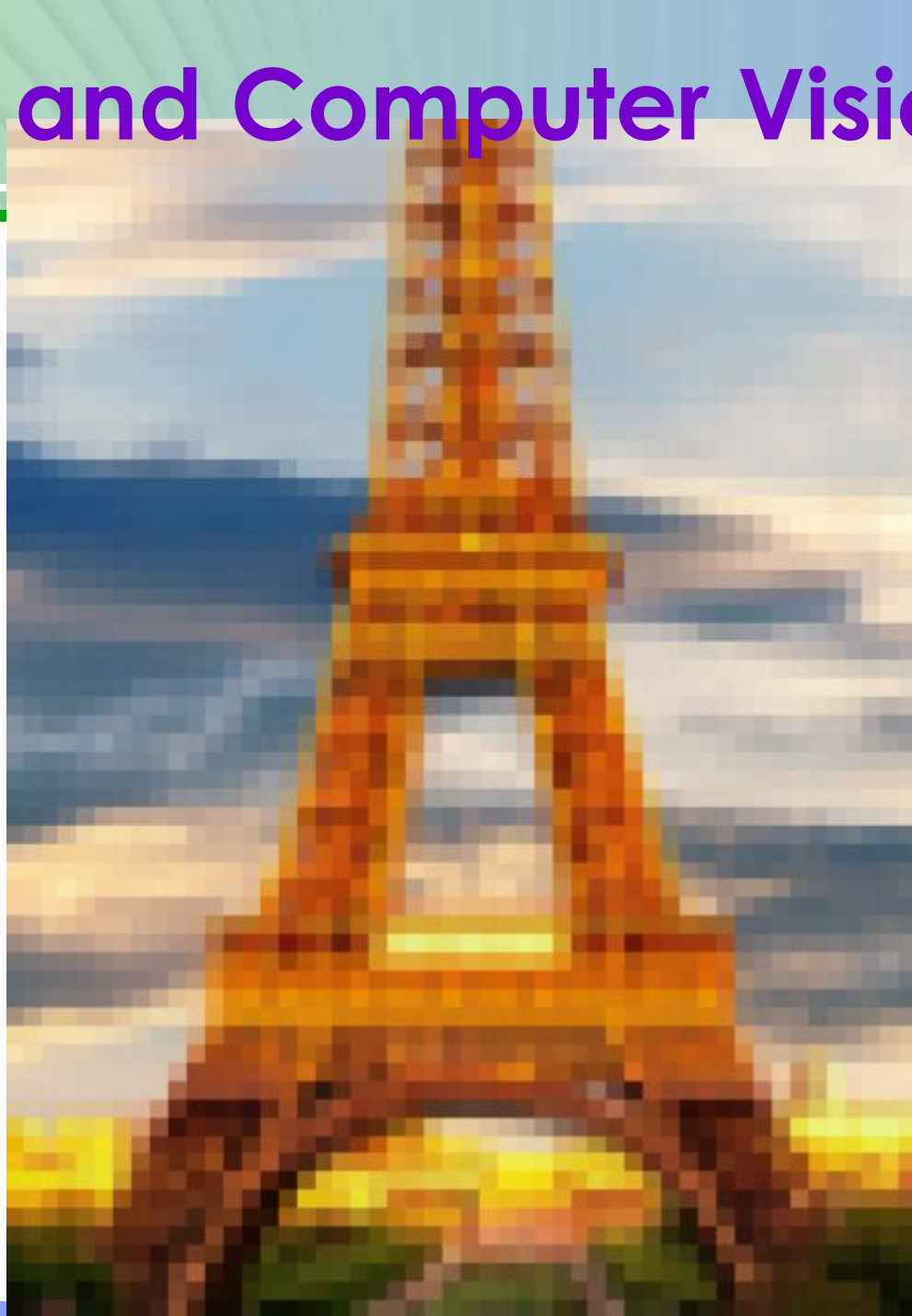
205	204	204	206	207
206	203	208	206	206
201	199	205	206	209
61	128	213	210	205
59	65	65	206	199

CNN and Computer Vision

<https://www.csfieldguide.org.nz/en/interactives/pixel-viewer/>



CNN and Computer Vision





CNN and Computer Vision

61	G 161	G 140	G 97	G 103	G 68	G 89	G 128	G 129	G 157	G 160	G 165	G 167	G 164	G 168	G 139	G 108	G 79
68	B 134	B 39	B 31	B 21	B 10	B 11	B 14	B 93	B 155	B 159	B 164	B 166	B 164	B 165	B 97	B 12	B 12
02	R 195	R 179	R 144	R 158	R 153	R 200	R 211	R 145	R 123	R 114	R 139	R 126	R 127	R 136	R 146	R 201	R 159
67	G 156	G 85	G 54	G 59	G 51	G 90	G 125	G 116	G 122	G 118	G 130	G 123	G 125	G 132	G 117	G 110	G 59
61	B 104	B 15	B 22	B 16	B 10	B 13	B 23	B 93	B 129	B 126	B 132	B 129	B 131	B 135	B 100	B 16	B 18
02	R 199	R 187	R 174	R 201	R 176	R 164	R 208	R 153	R 140	R 123	R 155	R 169	R 193	R 187	R 165	R 184	R 154
67	G 148	G 100	G 73	G 83	G 65	G 75	G 125	G 120	G 128	G 116	G 135	G 146	G 164	G 160	G 134	G 92	G 47
01	B 69	B 45	B 18	B 14	B 14	B 8	B 31	B 100	B 125	B 121	B 127	B 132	B 140	B 140	B 116	B 15	B 11
21	R 216	R 182	R 204	R 198	R 152	R 175	R 208	R 197	R 190	R 183	R 193	R 228	R 203	R 238	R 220	R 181	R 130
05	G 139	G 95	G 94	G 81	G 53	G 95	G 125	G 155	G 161	G 157	G 163	G 191	G 207	G 201	G 182	G 105	G 44
01	B 43	B 54	B 17	B 14	B 24	B 12	B 49	B 122	B 138	B 137	B 139	B 151	B 157	B 155	B 142	B 34	B 11
01	R 225	R 208	R 208	R 199	R 179	R 213	R 221	R 242	R 236	R 232	R 214	R 237	R 233	R 233	R 214	R 195	R 192
07	G 130	G 100	G 92	G 77	G 70	G 127	G 146	G 205	G 196	G 198	G 179	G 201	G 195	G 196	G 177	G 125	G 91
04	B 29	B 35	B 20	B 14	B 16	B 10	B 69	B 151	B 149	B 153	B 142	B 153	B 148	B 151	B 137	B 52	B 12
05	R 190	R 181	R 174	R 178	R 206	R 220	R 197	R 192	R 225	R 239	R 226	R 241	R 241	R 222	R 213	R 200	R 203
67	G 82	G 74	G 67	G 62	G 92	G 126	G 130	G 160	G 185	G 200	G 189	G 207	G 209	G 187	G 179	G 138	G 103
2	B 16	B 16	B 12	B 10	B 13	B 14	B 69	B 132	B 142	B 150	B 147	B 156	B 159	B 147	B 142	B 73	B 13
02	R 173	R 184	R 199	R 167	R 170	R 219	R 215	R 218	R 229	R 234	R 234	R 233	R 236	R 232	R 229	R 211	R 185
08	G 82	G 70	G 76	G 62	G 87	G 109	G 122	G 160	G 174	G 186	G 184	G 181	G 184	G 177	G 162	G 130	G 83
2	B 41	B 16	B 13	B 11	B 13	B 18	B 67	B 114	B 122	B 129	B 120	B 120	B 124	B 121	B 115	B 74	B 12
08	R 171	R 183	R 184	R 143	R 183	R 195	R 217	R 215	R 217	R 218	R 216	R 218	R 218	R 217	R 218	R 207	R 179
03	G 74	G 94	G 100	G 76	G 104	G 118	G 161	G 170	G 167	G 175	G 172	G 167	G 170	G 171	G 172	G 139	G 103
01	B 21	B 19	B 22	B 24	B 23	B 27	B 48	B 57	B 51	B 53	B 58	B 53	B 57	B 57	B 54	B 29	B 22
69	R 120	R 145	R 158	R 103	R 159	R 206	R 234	R 233	R 234	R 235	R 235	R 234	R 235	R 235	R 234	R 216	R 166
6	G 25	G 43	G 61	G 30	G 62	G 145	G 236	G 238	G 229	G 238	G 239	G 238	G 235	G 241	G 234	G 163	G 70
2	B 9	B 12	B 17	B 13	B 18	B 42	B 111	B 132	B 111	B 119	B 132	B 132	B 125	B 134	B 134	B 41	B 14
01	R 153	R 190	R 201	R 179	R 183	R 220	R 229	R 230	R 227	R 230	R 230	R 229	R 231	R 233	R 227	R 222	R 200
8	G 67	G 87	G 101	G 82	G 89	G 145	G 167	G 171	G 162	G 162	G 172	G 169	G 169	G 176	G 168	G 140	G 98
7	B 8	B 12	B 13	B 9	B 12	B 26	B 49	B 55	B 45	B 47	B 53	B 46	B 46	B 55	B 50	B 20	B 11
00	R 214	R 207	R 213	R 203	R 222	R 205	R 212	R 214	R 197	R 226	R 203	R 212	R 217	R 198	R 222	R 204	R 210
8	G 101	G 98	G 103	G 92	G 110	G 97	G 103	G 102	G 93	G 106	G 95	G 98	G 100	G 92	G 102	G 94	G 97
2	B 8	B 9	B 10	B 8	B 8	B 9	B 10	B 9	B 12	B 5	B 13	B 8	B 7	B 11	B 9	B 9	B 8
03	R 231	R 220	R 223	R 216	R 232	R 218	R 225	R 225	R 212	R 233	R 214	R 222	R 226	R 210	R 233	R 216	R 219
09	G 140	G 129	G 132	G 123	G 139	G 125	G 135	G 132	G 121	G 141	G 120	G 127	G 131	G 113	G 137	G 119	G 129
6	B 7	B 8	B 9	B 9	B 7	B 8	B 11	B 8	B 11	B 3	B 14	B 6	B 7	B 12	B 7	B 10	B 8
08	R 212	R 223	R 224	R 212	R 212	R 208	R 212	R 211	R 216	R 206	R 212	R 210	R 210	R 211	R 210	R 209	R 208



R 176	R 164	R 208	R 153	R 140	R 123	R 155	R 169	R 193	R 187
G 65	G 75	G 125	G 120	G 128	G 116	G 135	G 146	G 164	G 160
B 14	B 8	B 31	B 100	B 125	B 121	B 127	B 132	B 140	B 140
R 152	R 175	R 208	R 197	R 190	R 183	R 193	R 228	R 243	R 238
G 53	G 95	G 125	G 155	G 161	G 157	G 163	G 191	G 207	G 201
B 24	B 12	B 49	B 122	B 138	B 137	B 139	B 151	B 157	B 155
R 179	R 213	R 221	R 242	R 236	R 232	R 214	R 237	R 233	R 233
G 70	G 127	G 146	G 205	G 196	G 198	G 179	G 201	G 195	G 196
B 16	B 10	B 69	B 151	B 149	B 153	B 142	B 153	B 148	B 151
R 206	R 220	R 197	R 192	R 225	R 239	R 226	R 241	R 241	R 222
G 92	G 126	G 130	G 160	G 185	G 200	G 189	G 207	G 209	G 187
B 13	B 14	B 69	B 132	B 142	B 150	B 147	B 156	B 159	B 147
R 170	R 219	R 215	R 218	R 229	R 234	R 234	R 233	R 236	R 232
G 87	G 109	G 122	G 160	G 174	G 186	G 184	G 181	G 184	G 177
B 13	B 18	B 67	B 114	B 122	B 129	B 120	B 120	B 124	B 121
R 183	R 195	R 217	R 215	R 217	R 218	R 216	R 218	R 218	R 217
G 104	G 118	G 161	G 170	G 167	G 175	G 172	G 167	G 170	G 171
B 23	B 27	B 48	B 57	B 51	B 53	B 58	B 53	B 57	B 57
R 159	R 206	R 234	R 233	R 234	R 233	R 235	R 234	R 235	R 235
G 62	G 145	G 226	G 238	G 229	G 228	G 239	G 233	G 235	G 241
B 18	B 42	B 111	B 132	B 113	B 119	B 137	B 118	B 125	B 133







G 243	G 244	G 238	G 190	G 125	G 141	G 108	G 199	G 130	G 110	G 57	G 157	G 194	G 244	G 241
B 243	B 244	B 238	B 190	B 125	B 141	B 108	B 199	B 130	B 110	B 57	B 157	B 194	B 244	B 241
R 244	R 244	R 231	R 203	R 120	R 73	R 126	R 102	R 93	R 116	R 134	R 135	R 117	R 241	R 241
G 244	G 244	G 231	G 203	G 120	G 73	G 126	G 102	G 93	G 116	G 134	G 135	G 117	G 241	G 241
B 244	B 244	B 231	B 203	B 120	B 73	B 126	B 102	B 93	B 116	B 134	B 135	B 117	B 241	B 241
R 245	R 244	R 200	R 147	R 165	R 136	R 59	R 134	R 143	R 108	R 66	R 128	R 138	R 236	R 240
G 245	G 244	G 200	G 147	G 165	G 136	G 59	G 134	G 143	G 108	G 66	G 128	G 138	G 236	G 240
B 245	B 244	B 200	B 147	B 165	B 136	B 59	B 134	B 143	B 108	B 66	B 128	B 138	B 236	B 240
R 244	R 245	R 179	R 204	R 139	R 78	R 147	R 235	R 120	R 121	R 86	R 127	R 91	R 225	R 238
G 244	G 245	G 179	G 204	G 139	G 78	G 147	G 235	G 120	G 121	G 86	G 127	G 91	G 225	G 238
B 244	B 245	B 179	B 204	B 139	B 78	B 147	B 235	B 120	B 121	B 86	B 127	B 91	B 225	B 238
R 245	R 240	R 181	R 189	R 119	R 140	R 70	R 227	R 42	R 110	R 100	R 127	R 164	R 200	R 238
G 245	G 240	G 181	G 189	G 119	G 141	G 70	G 227	G 42	G 110	G 100	G 127	G 164	G 200	G 238
B 245	B 240	B 181	B 189	B 119	B 140	B 70	B 227	B 42	B 110	B 100	B 127	B 164	B 200	B 238
R 246	R 231	R 195	R 162	R 138	R 108	R 133	R 186	R 135	R 95	R 83	R 117	R 199	R 174	R 245
G 246	G 231	G 195	G 162	G 138	G 108	G 133	G 186	G 135	G 95	G 83	G 117	G 199	G 174	G 245
B 246	B 231	B 195	B 162	B 138	B 108	B 133	B 186	B 135	B 95	B 83	B 117	B 199	B 174	B 245
R 247	R 224	R 200	R 211	R 172	R 130	R 221	R 190	R 120	R 110	R 171	R 184	R 143	R 157	R 243
G 247	G 224	G 200	G 211	G 172	G 130	G 221	G 190	G 120	G 110	G 171	G 184	G 143	G 157	G 243
B 247	B 224	B 200	B 211	B 172	B 130	B 221	B 190	B 120	B 110	B 171	B 184	B 143	B 157	B 243
R 248	R 172	R 205	R 213	R 176	R 111	R 175	R 174	R 160	R 69	R 162	R 88	R 218	R 196	R 236
G 248	G 172	G 205	G 213	G 176	G 111	G 175	G 174	G 160	G 69	G 162	G 88	G 218	G 196	G 236
B 248	B 172	B 205	B 213	B 176	B 111	B 175	B 174	B 160	B 69	B 162	B 88	B 218	B 196	B 236
R 241	R 189	R 172	R 129	R 184	R 163	R 198	R 134	R 205	R 114	R 186	R 122	R 185	R 150	R 197
G 241	G 189	G 172	G 129	G 184	G 163	G 198	G 134	G 205	G 114	G 186	G 122	G 185	G 150	G 197
B 241	B 189	B 172	B 129	B 184	B 163	B 198	B 134	B 205	B 114	B 186	B 122	B 185	B 150	B 197
R 223	R 211	R 214	R 86	R 167	R 153	R 202	R 161	R 162	R 139	R 85	R 158	R 154	R 156	R 157
G 223	G 211	G 214	G 86	G 167	G 153	G 202	G 161	G 162	G 139	G 85	G 158	G 154	G 156	G 157
B 223	B 211	B 214	B 86	B 167	B 153	B 202	B 161	B 162	B 139	B 85	B 158	B 154	B 156	B 157
R 197	R 189	R 185	R 108	R 170	R 117	R 141	R 157	R 119	R 84	R 171	R 173	R 173	R 209	R 175
G 197	G 189	G 185	G 108	G 170	G 117	G 141	G 157	G 119	G 84	G 171	G 173	G 173	G 209	G 175
B 197	B 189	B 185	B 108	B 170	B 117	B 141	B 157	B 119	B 84	B 171	B 173	B 173	B 209	B 175
R 201	R 194	R 183	R 116	R 235	R 208	R 161	R 185	R 226	R 155	R 130	R 117	R 192	R 180	R 136
G 201	G 194	G 183	G 116	G 235	G 208	G 161	G 185	G 226	G 155	G 130	G 117	G 192	G 180	G 136
B 201	B 194	B 183	B 116	B 235	B 208	B 161	B 185	B 226	B 155	B 130	B 117	B 192	B 180	B 136



Exploring Visual Processing Tasks: From CNNs to Computer Vision



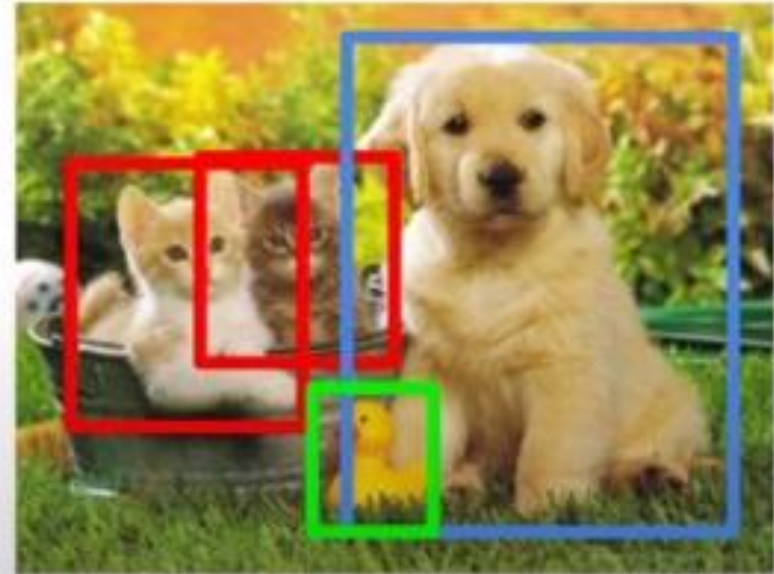
CNN and Computer Vision

Classification



CAT

Object Detection



CAT, DOG, DUCK



CNN and Computer Vision

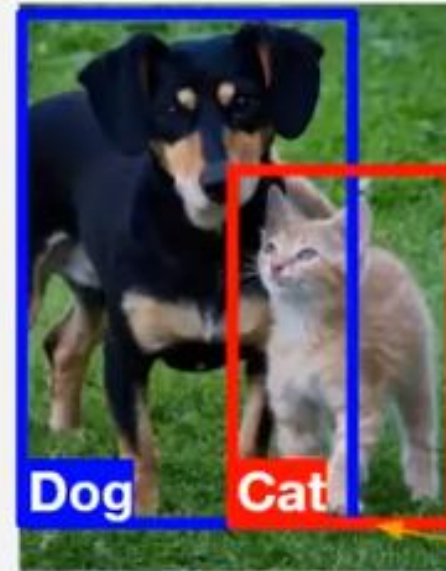
Classification

- One object and label per image



Object Detection

- Multiple objects per image
- Determine objects' location



Bounding Box

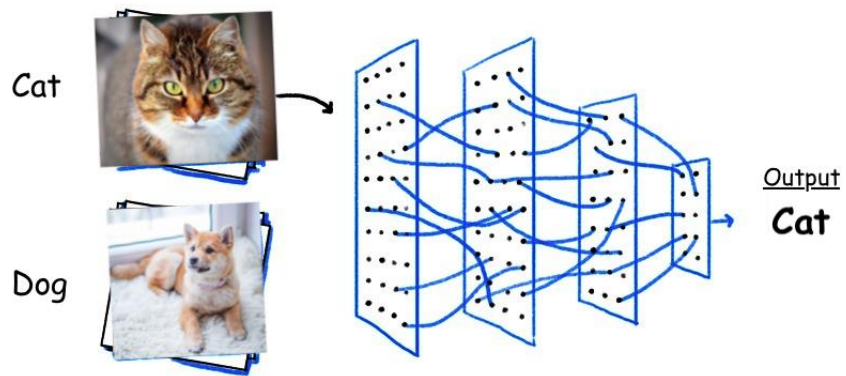
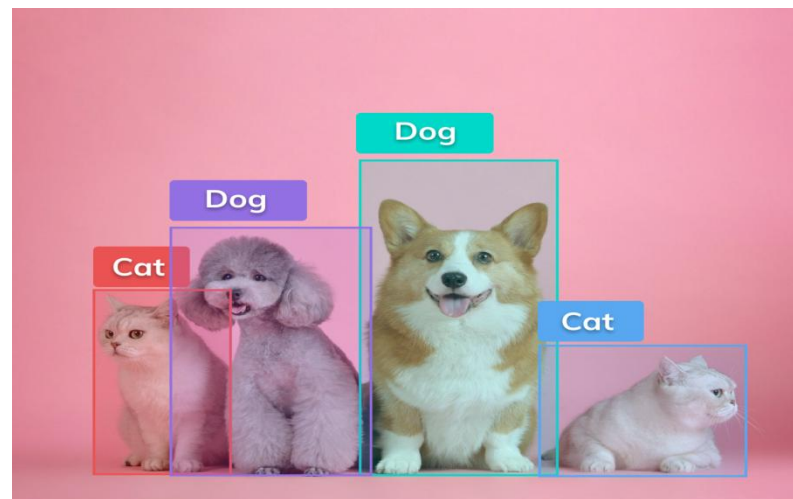
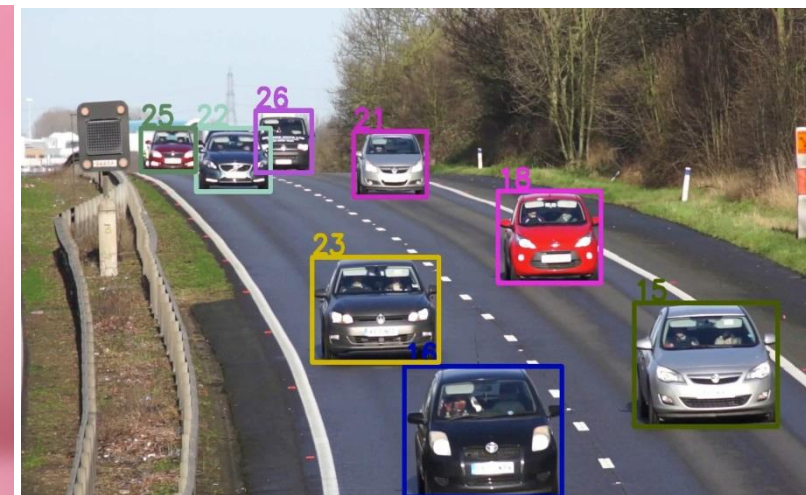


Image Classification



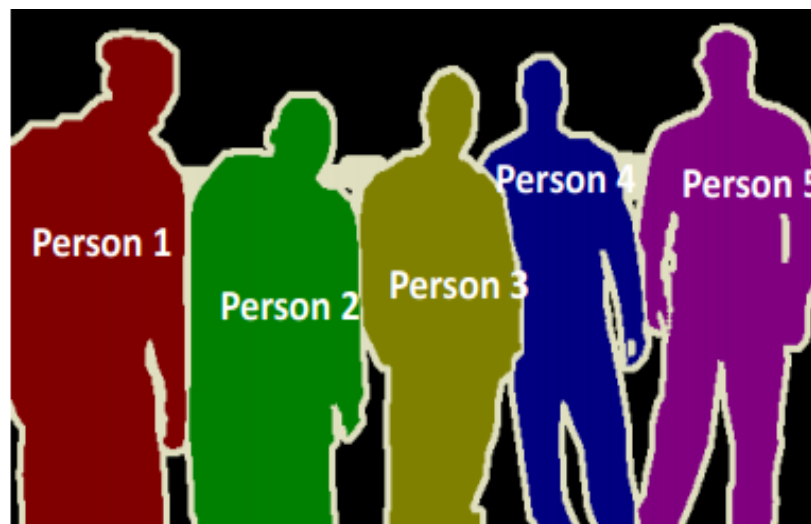
Object Detection



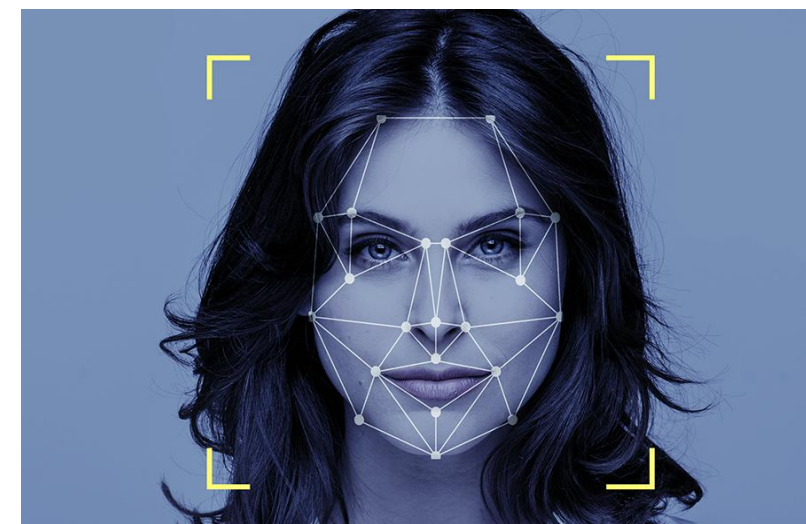
Object Tracking



Semantic Segmentation



Instance Segmentation



Face Detection and Recognition



CNN and Computer Vision

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

Classification + Localization



CAT

Single Object

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



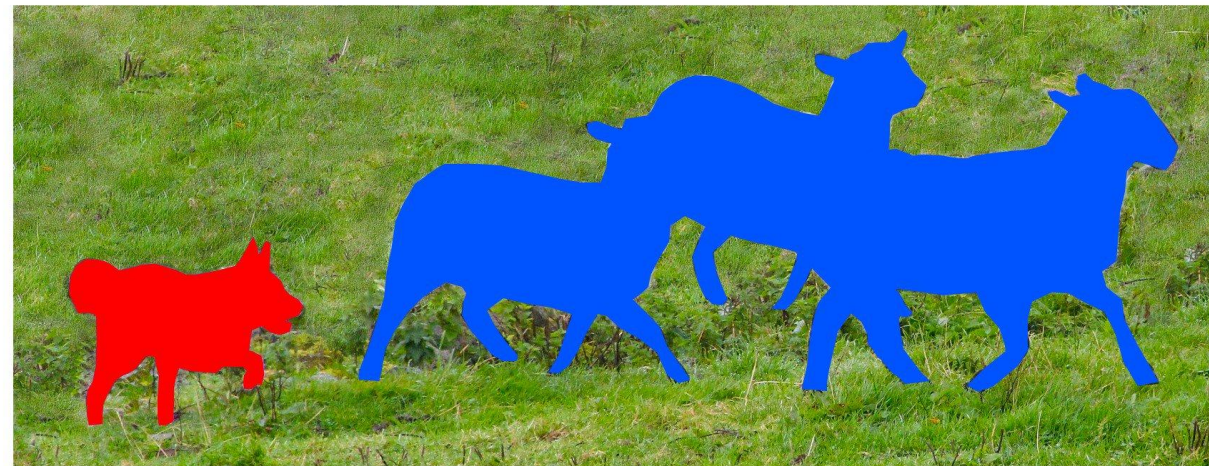
DOG, DOG, CAT



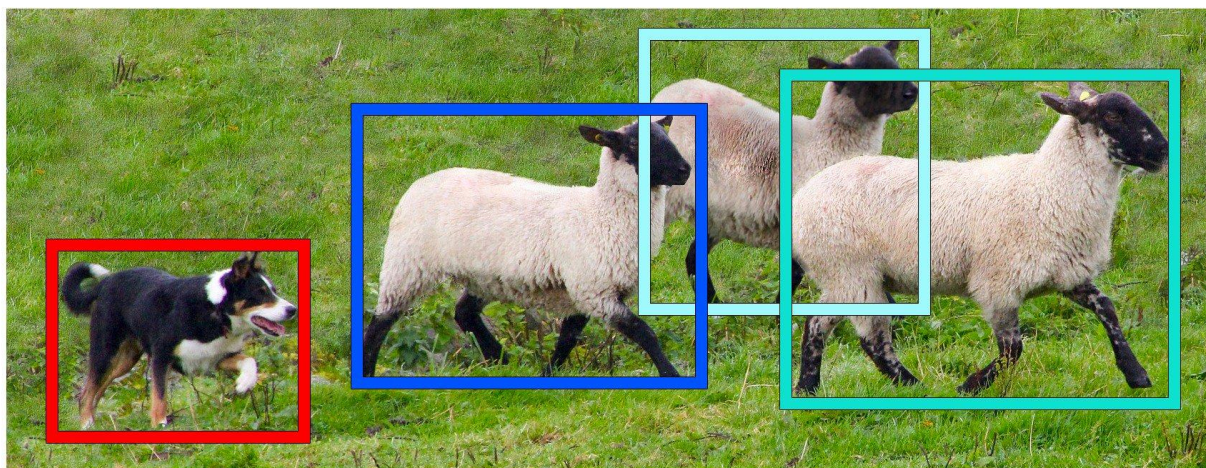
CNN and Computer Vision



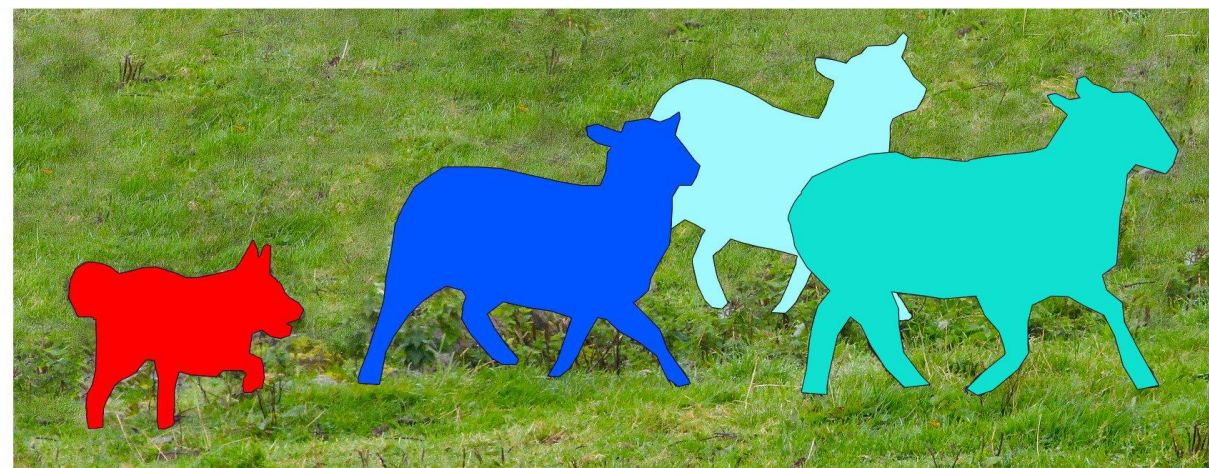
Image Recognition



Semantic Segmentation



Object Detection



Instance Segmentation



CNN and Computer Vision

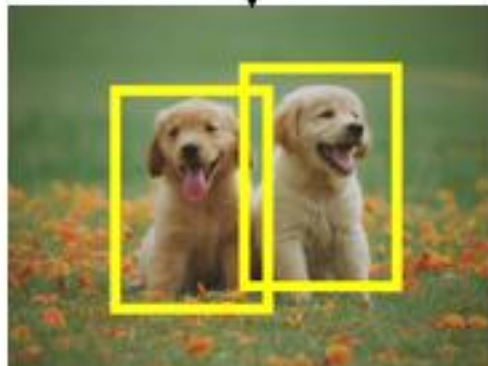
Input Image



Image Classification



Object Detection



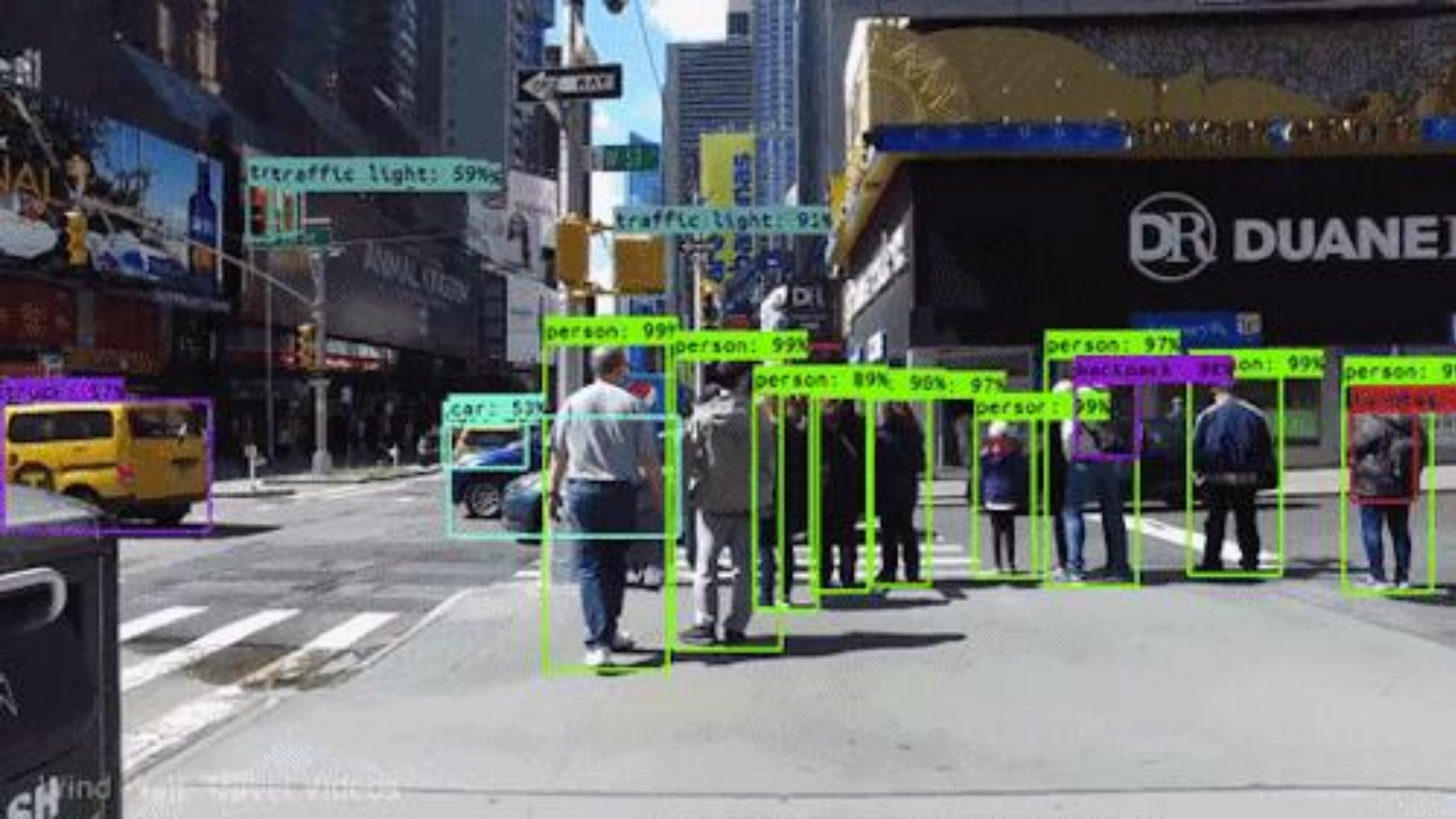
Instance Segmentation





OBJECT DETECTION





traffic light: 59%

traffic light: 91%

person: 99%

person: 99%

person: 97%

backpack: 98%

person: 97%

car: 53%

person: 89%

person: 98%

person: 97%

person: 99%

person: 99%

person: 97%



person

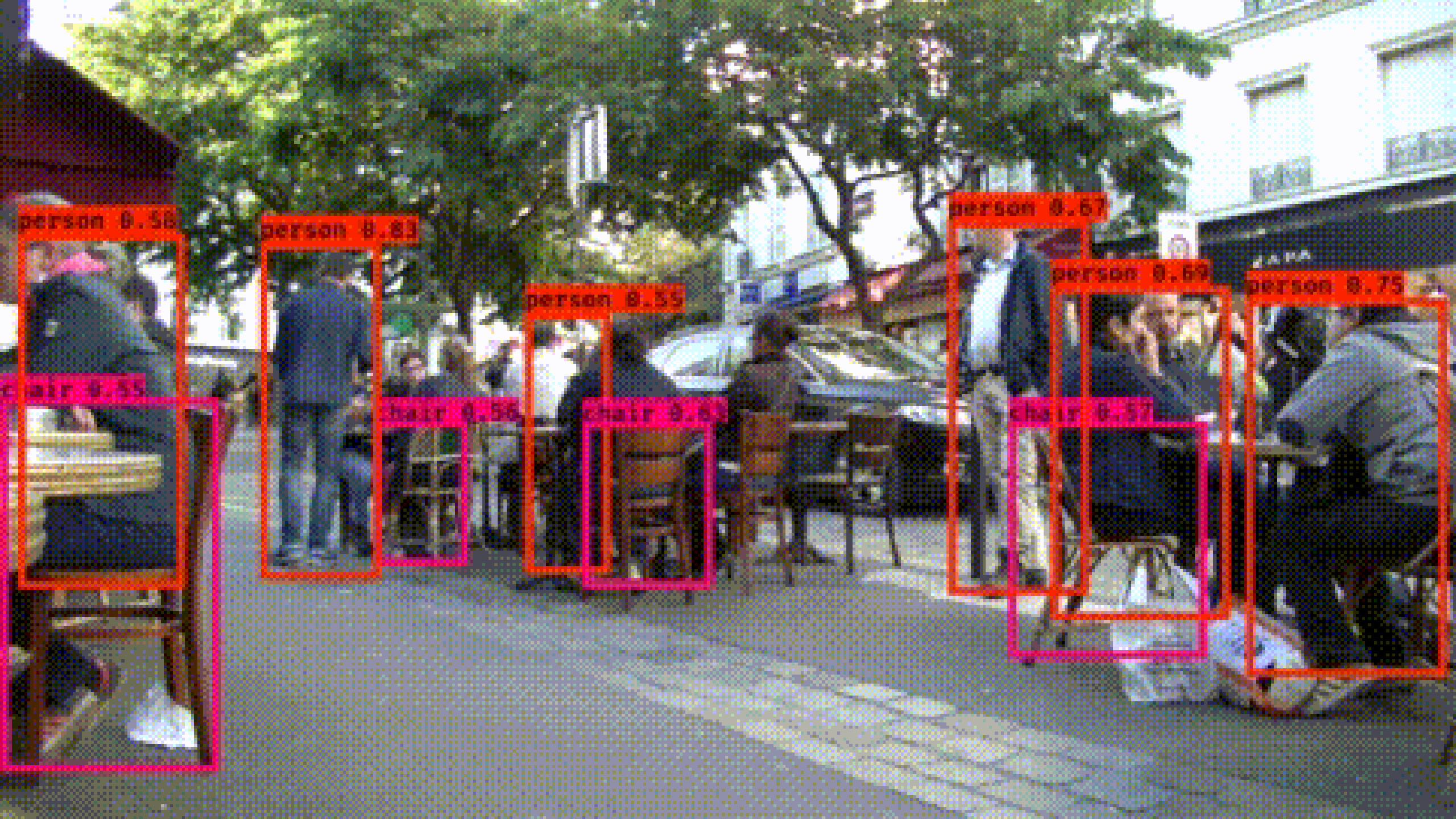
person

horse

person

person

gifs.com



person 0.58

person 0.83

person 0.35

person 0.67

person 0.69

person 0.75

chair 0.55

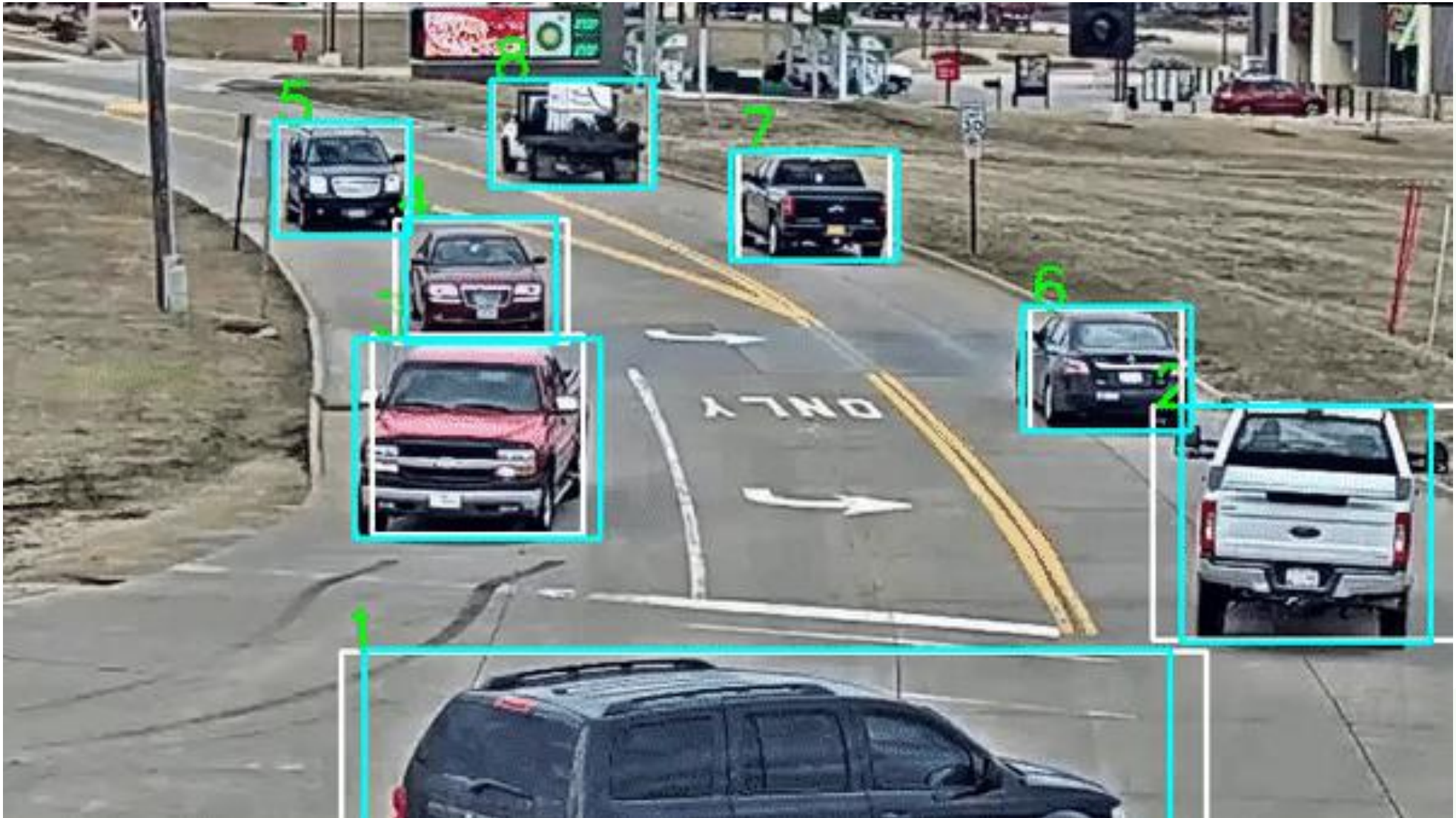
chair 0.56

chair 0.63

chair 0.57



OBJECT TRACKING





SEMANTIC SEGMENTATION



road
sidewalk
building
wall
fence
pole
traffic light
traffic sign
vegetation
terrain
sky
person
rider
car
truck
bus
train
motorcycle
bicycle



FACE DETECTION AND RECOGNITION





FACE DETECTION AND RECOGNITION

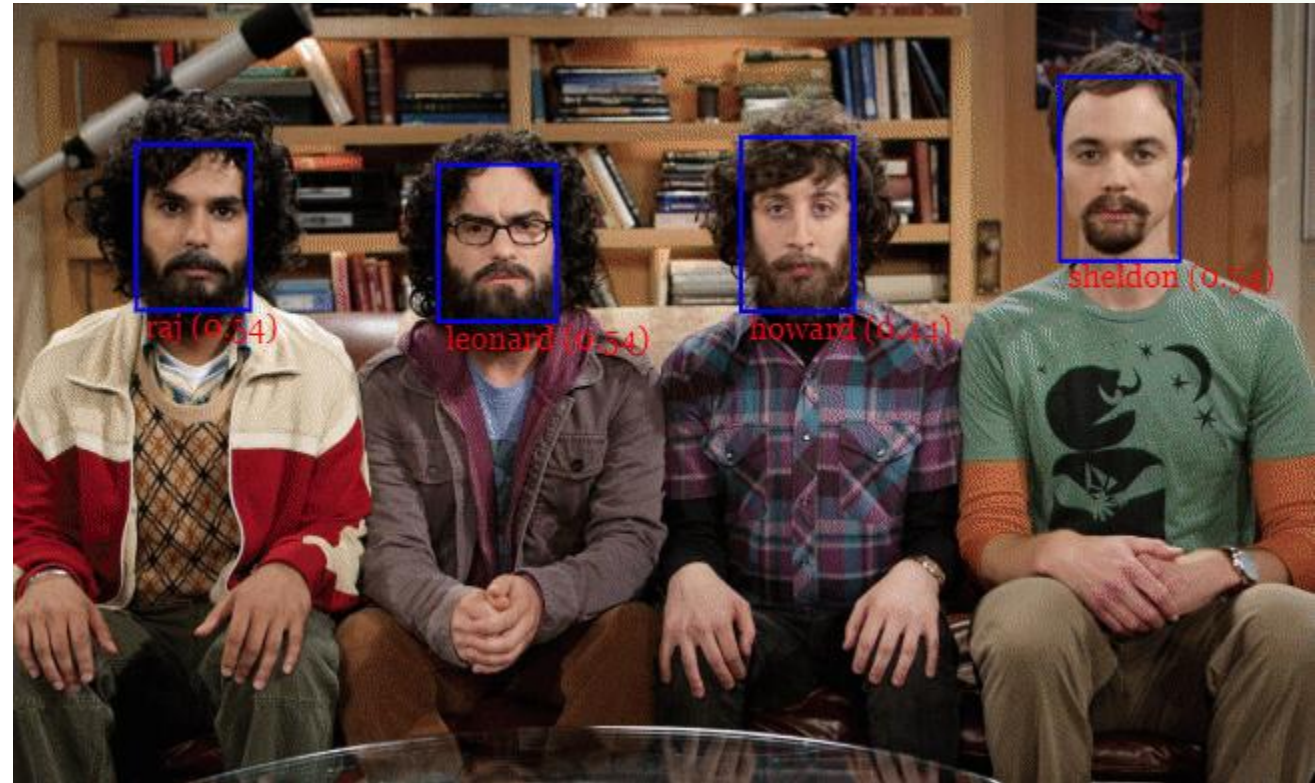
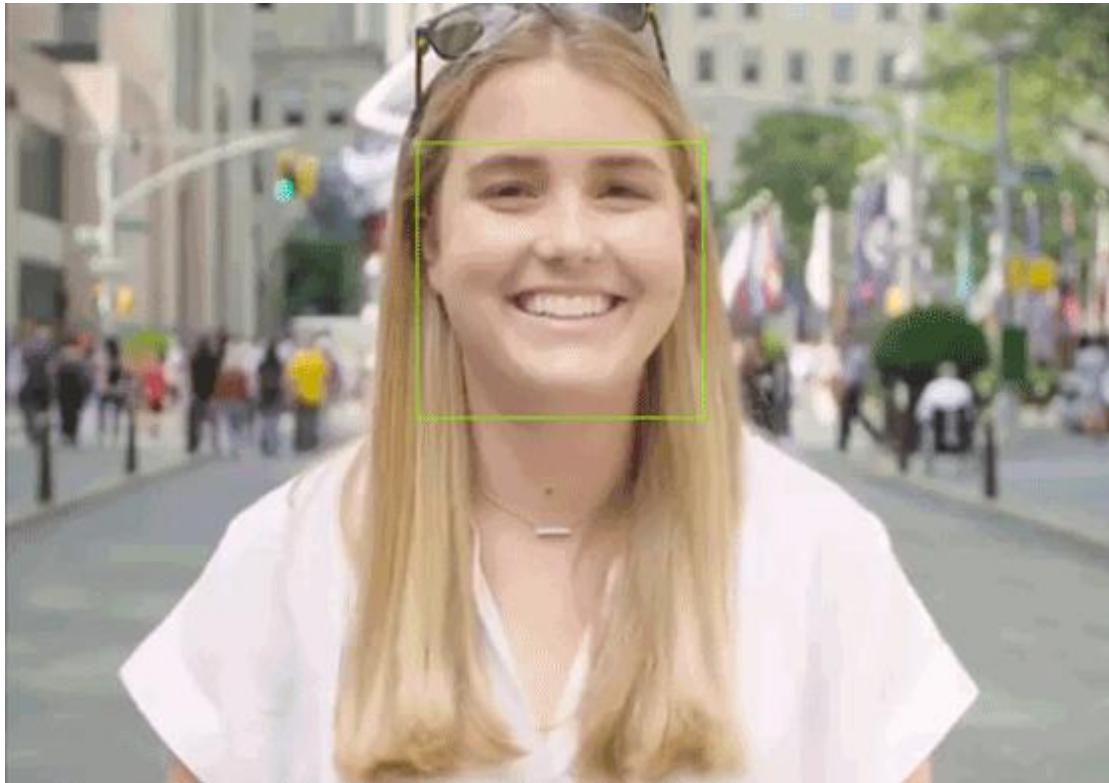




IMAGE PROCESSING

A word cloud of technology-related terms is displayed on a light blue background. The words are in various sizes and colors (black, blue, green). A hand holding a white marker is visible on the right side, pointing towards the word cloud. The words include: technology, image, processing, digital, computer, internet, data, design, website, nature, future, people, process, innovation, advertising, graphic, media, art, share, web, element, layout, success, devices, concepts, software, work, shape, laptop, exploration, choice, modern, responsibility, travel, options, information, presentation, business, job, employment, casual, search, engineering, reusable, label, tech, creative, professional, environment, dramatic, color, social, inspiration, world, chart, step, exchange, and solution.

technology
image
processing
digital
computer
internet
data
design
website
nature
future
people
process
innovation
advertising
graphic
media
art
share
web
element
layout
success
devices
concepts
software
work
shape
laptop
exploration
choice
modern
responsibility
travel
options
information
presentation
business
job
employment
casual
search
engineering
reusable
label
tech
creative
professional
environment
dramatic
color
social
inspiration
world
chart
step
exchange
solution

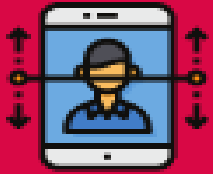


IMAGE PROCESSING

1

**Process an Image to
generate another image**

2

Input and output is image

3



Input [image]

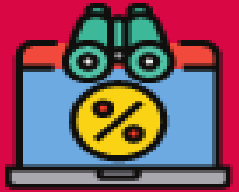


output [image]

4

**Smoothing, sharpening, compression,
contrast, watermarking brightness, edges,
region.**

COMPUTER VISION

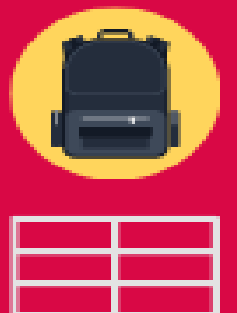


**Process an Image to produce
desired result in real time**

**Input as an image
Output as desired result**



Input [image]



**Position, Action, Identification,
Measure, Label, Projection**

Image Processing

Focuses on processing the image.
Both input and output are images.



SMOOTHING &
SHARPENING

CHANGE
CONTRAST &
BRIGHTNESS

HIGHLIGHT EDGES
& REGIONS

WATERMARKING

COMPRESSION

CALIBRATION

VS

Computer Vision

Focuses on making sense of what a
machine sees. The system inputs
an image and outputs task-specific
knowledge.



LABELLING

POSITION

IDENTIFICATION

MEASUREMENT

ACTION

PROJECTION



Senior AI/ML Engineer - Computer Vision Specialty

- Experience working in a specialized Computer Vision role within the last 2 years required.
- Experience with working in a collaborative CI/CD software development environment, including use of git.
- Experience with developing deep learning models for diverse computer vision tasks (e.g., segmentation, recognition, classification, domain adaptation) using PyTorch, Tensorflow, or other deep learning frameworks.
- Experience in machine learning, software engineering, and Python.
- Experience with one or more open-source computer vision libraries such as Pillow, scikit-image, and OpenCV.

Analyze, design, code, and test multiple components of application code across one or more clients for enterprise-level identity management software. Perform maintenance, enhancements and/or development work. Must be proficient developing software with Python programming language, experience developing algorithms and software on Linux and Windows, and familiar with image processing concepts, file formats, and tools and python packages like PIL/Pillow and Scikit-Image

Required: PyTorch, Tensorflow, Keras, Numpy, OpenCV/Pillow.
Additionally Preferred: Scikit-Learn, Pandas, etc.



OpenCV and Image Processing Engineer

Computer Vision Engineer

Requirements:

- Strong experience with OpenCV and image processing algorithms
 - Proficient in Python
 - Experience with image processing libraries such as OpenCV, Scikit-image, scikit-learn, Tensorflow, Pytorch, Keras.
 - Strong understanding of computer vision concepts such as feature detection, object recognition, and image segmentation
-
- Passion or Experience in Computer Vision / Machine Learning technologies and tools
 - Experience with computer vision tools like OpenCV, YOLO, PIP, Tensorflow, PyTorch . Experience with Deep Learning, Neural Networks and Artificial Intelligence
 - Technical understanding of computer vision and machine learning
 - Familiar with state-of-the-art CV/ML models and tools